

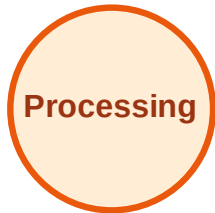
BFS Traversal

Step 1: Start at A

Legend



Complete



Processing



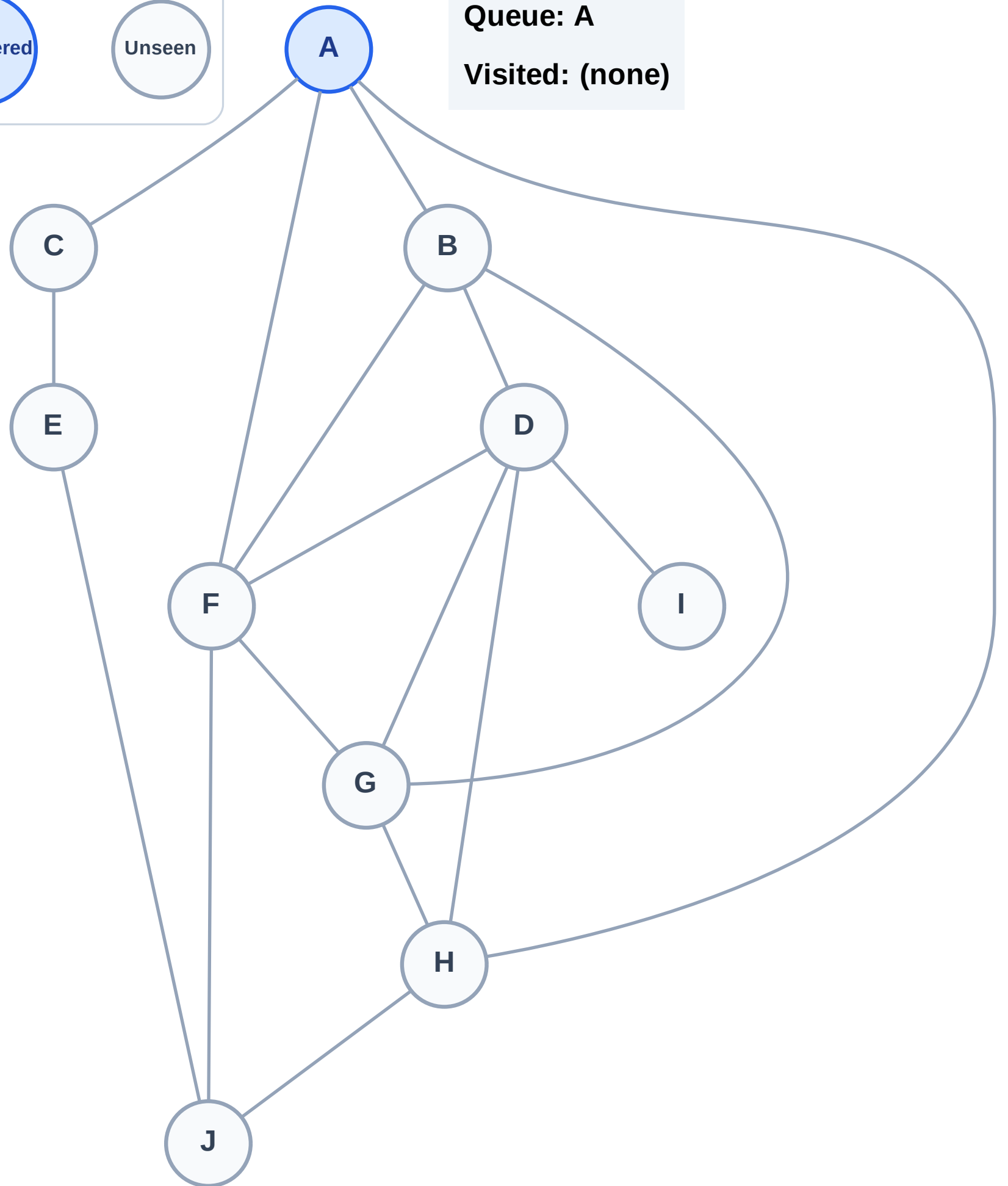
Discovered



Unseen

Queue: A

Visited: (none)



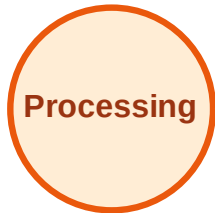
BFS Traversal

Step 2: Process node A

Legend



Complete



Processing



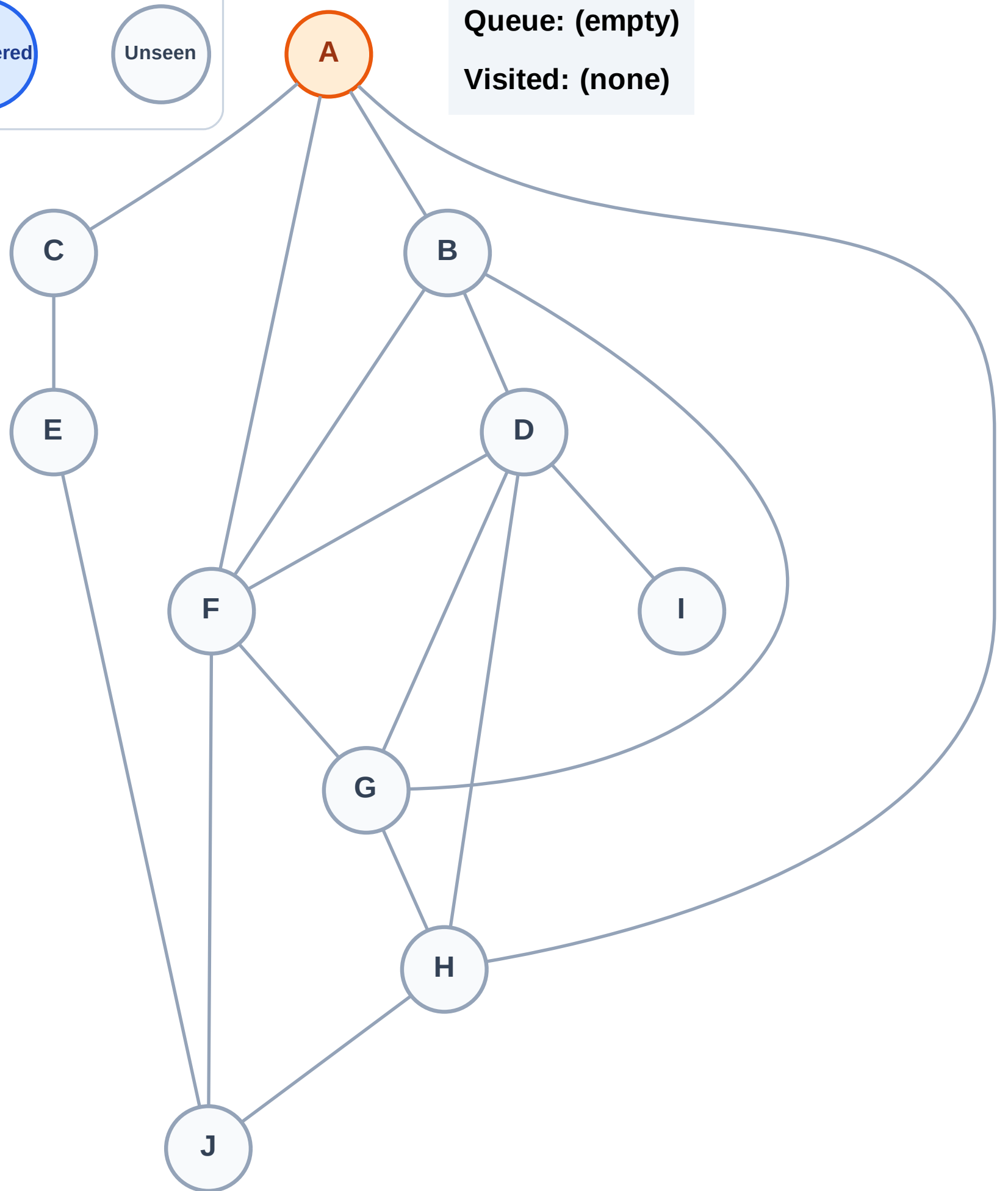
Discovered



Unseen

Queue: (empty)

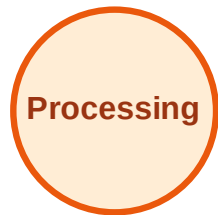
Visited: (none)



BFS Traversal

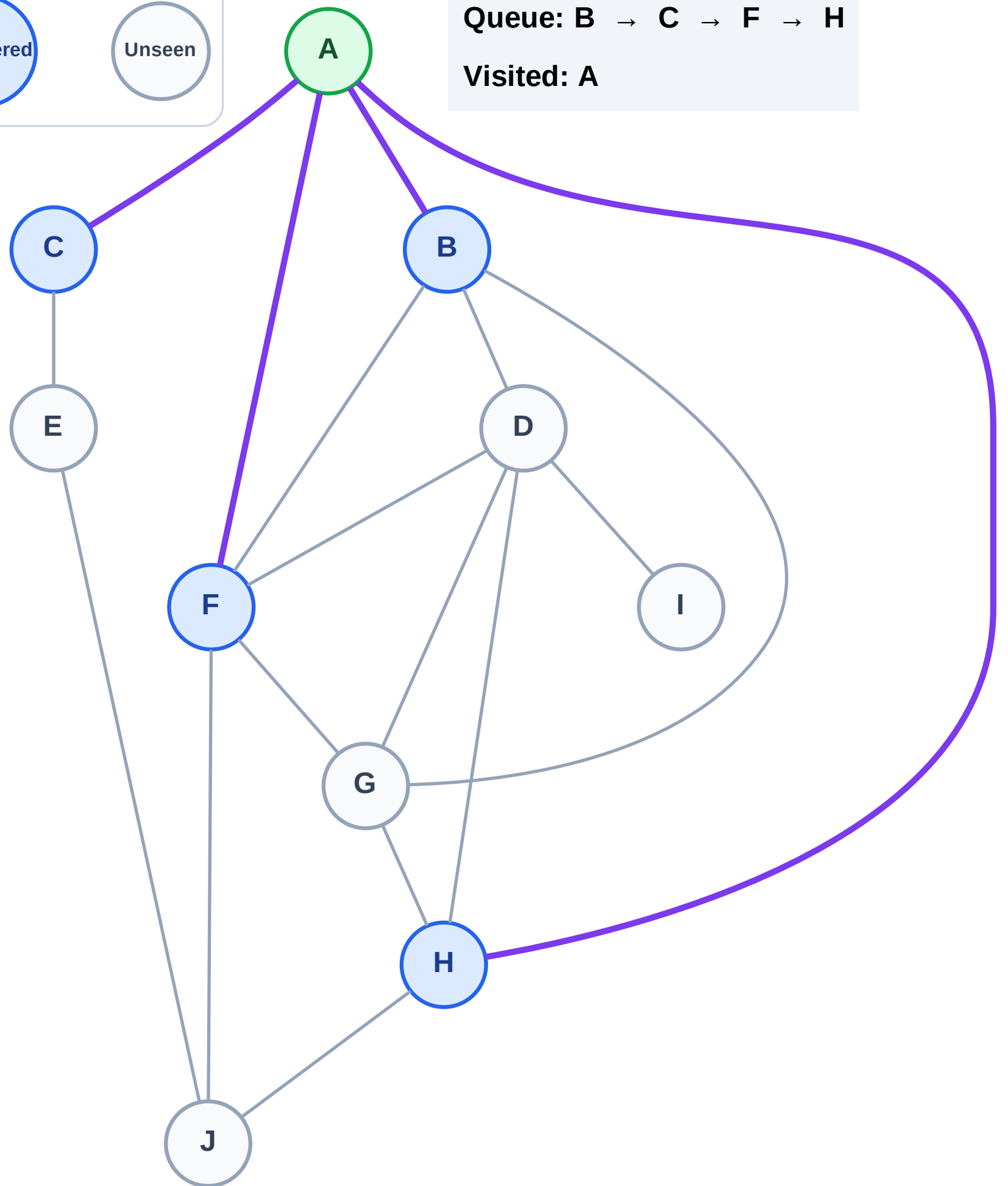
Step 3: Discover B, C, F, H from A

Legend



Queue: B → C → F → H

Visited: A



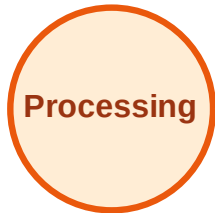
BFS Traversal

Step 4: Process node B

Legend



Complete



Processing



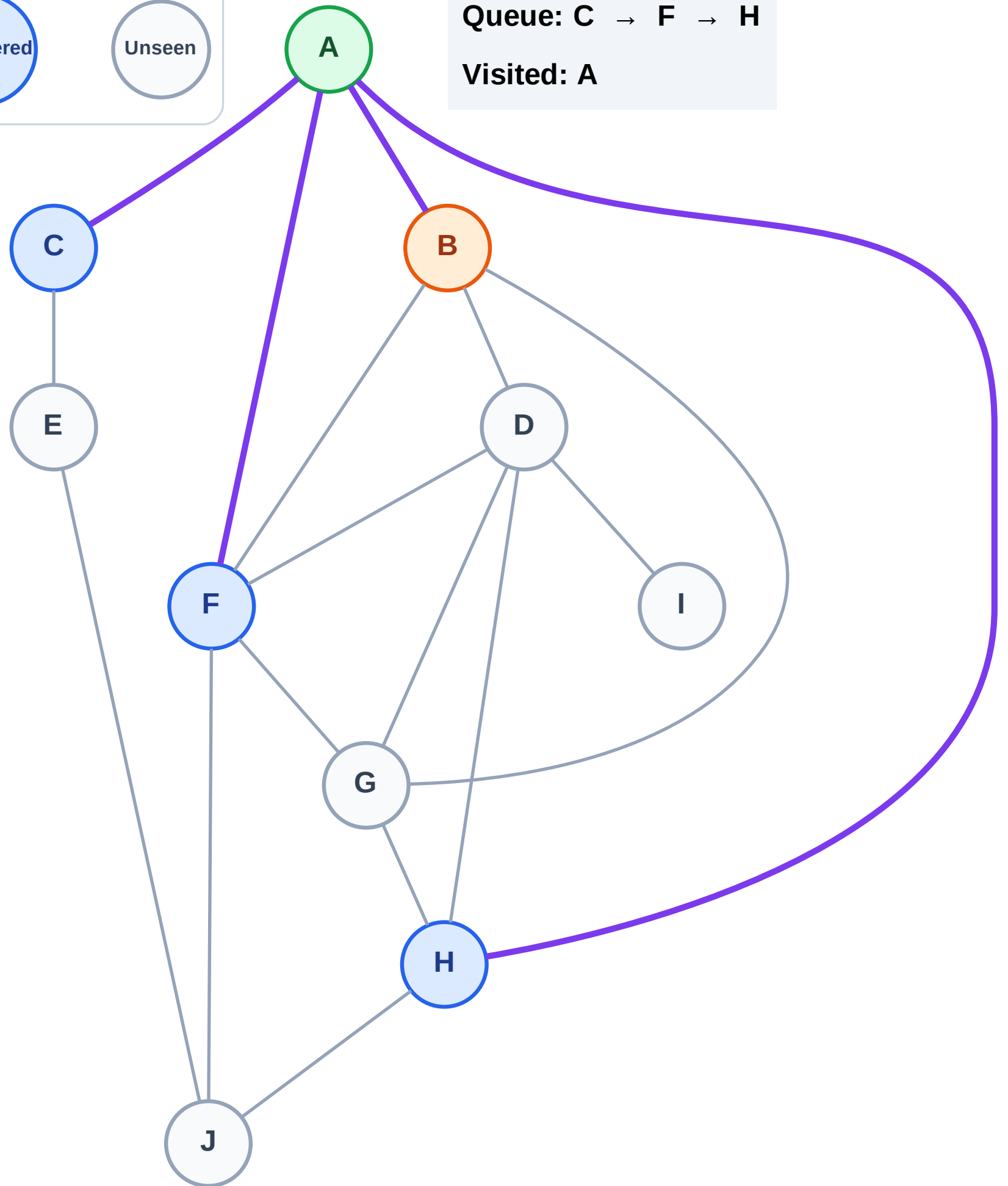
Discovered



Unseen

Queue: C → F → H

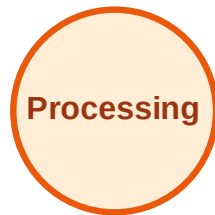
Visited: A



BFS Traversal

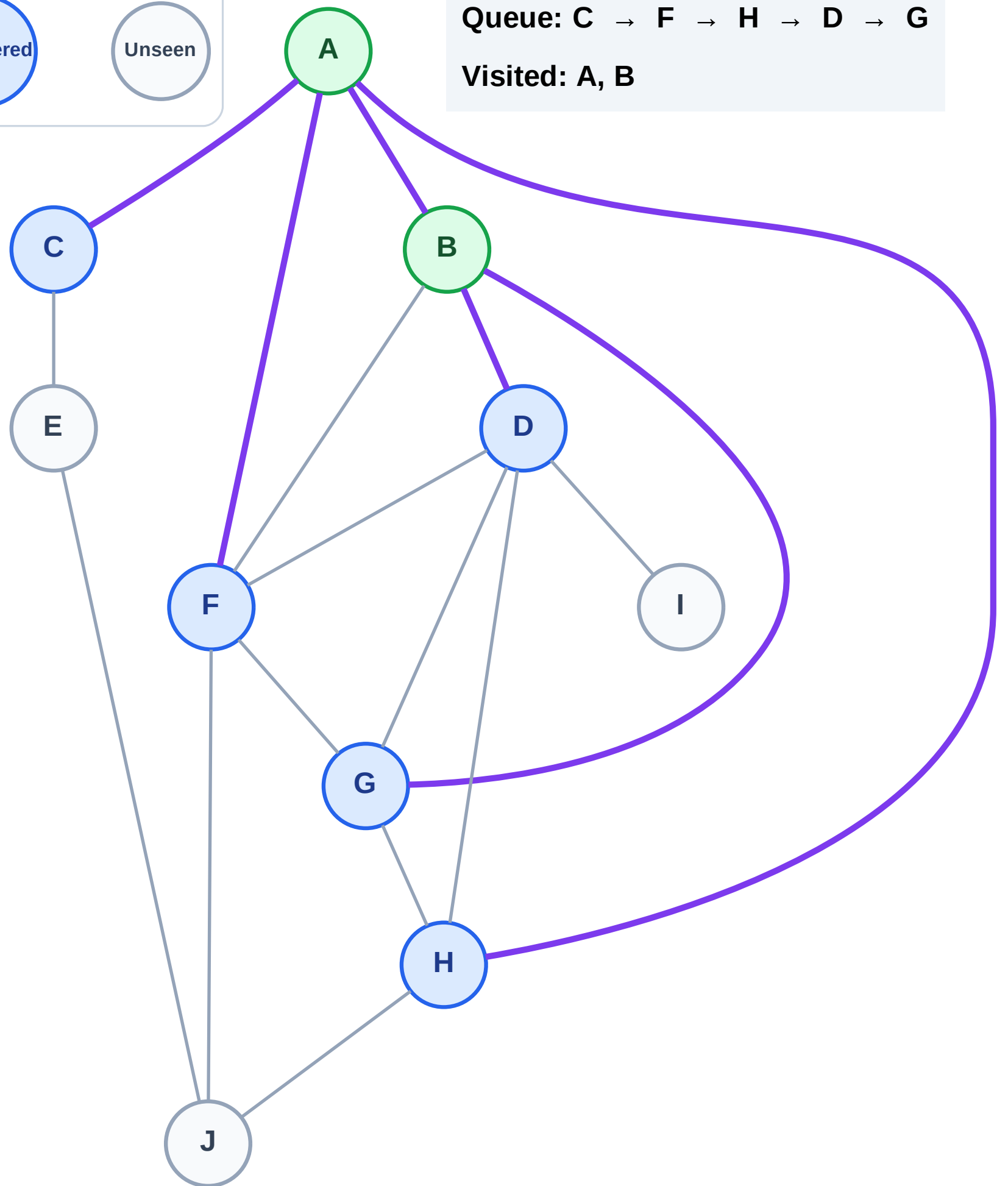
Step 5: Discover D, G from B

Legend



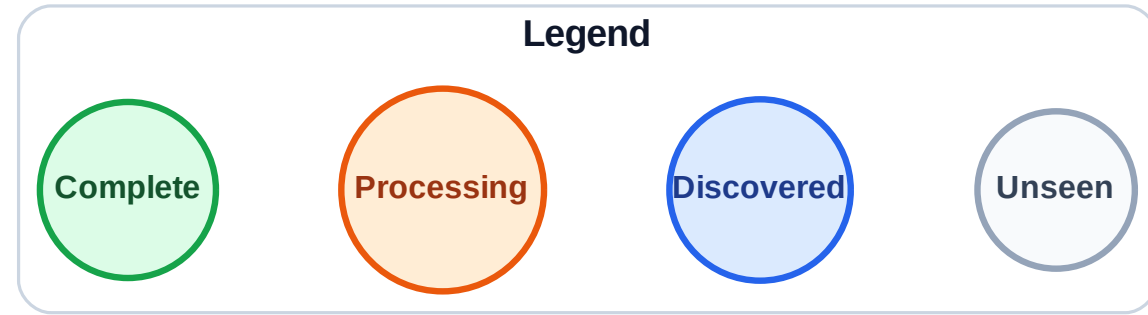
Queue: C → F → H → D → G

Visited: A, B



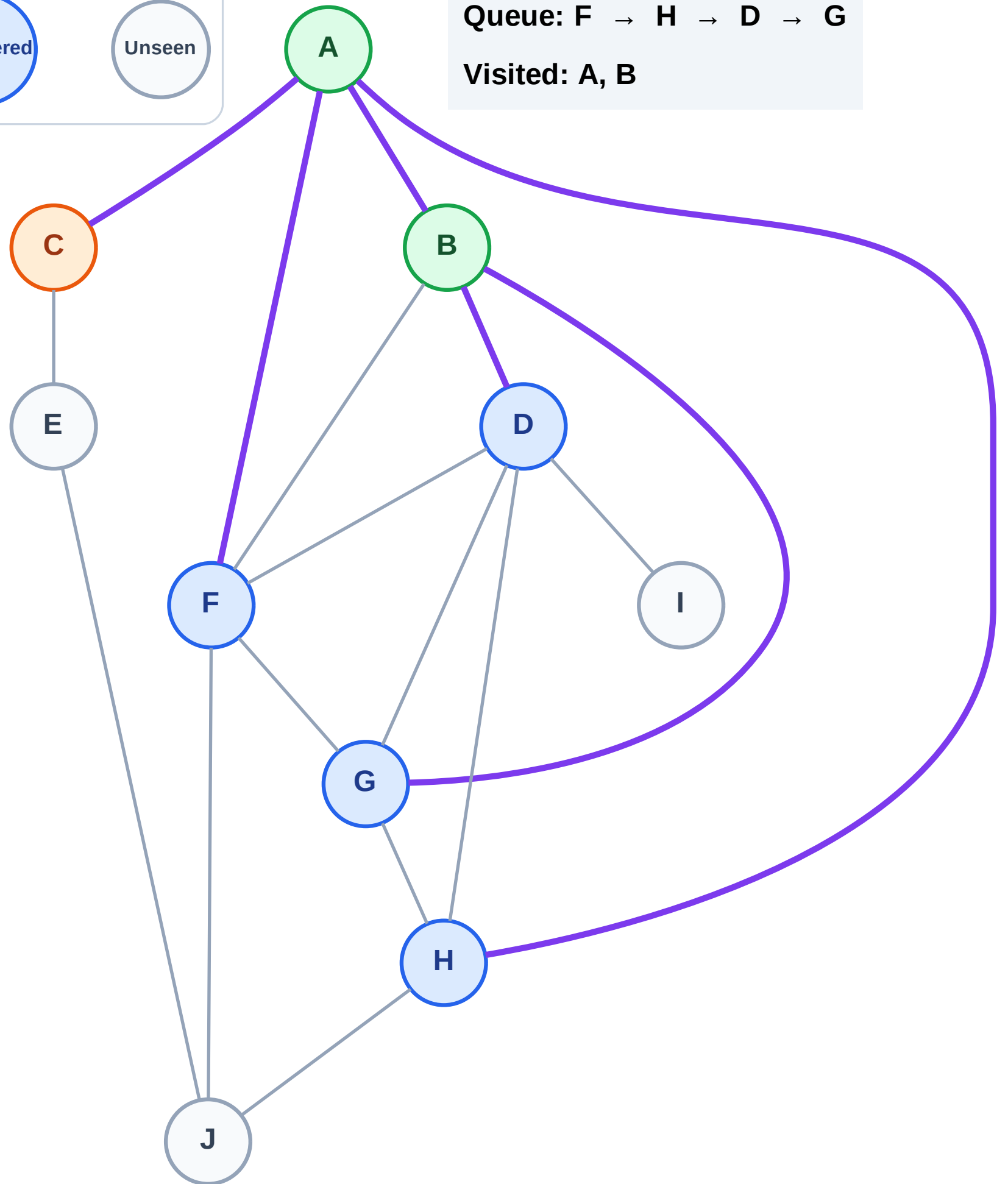
BFS Traversal

Step 6: Process node C



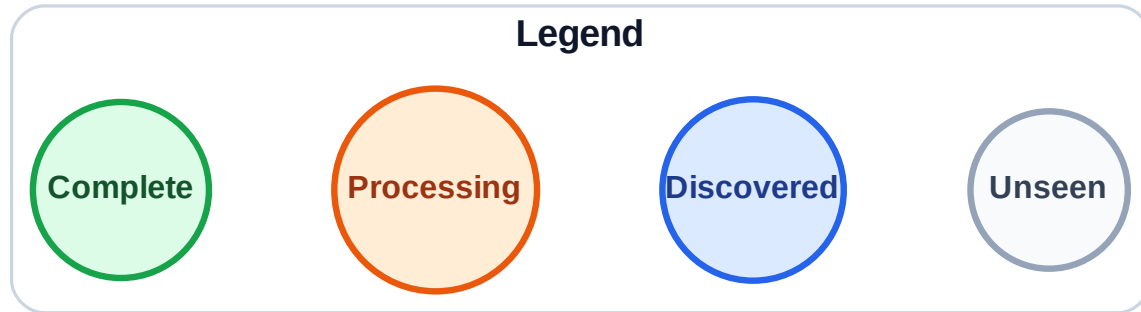
Queue: F → H → D → G

Visited: A, B

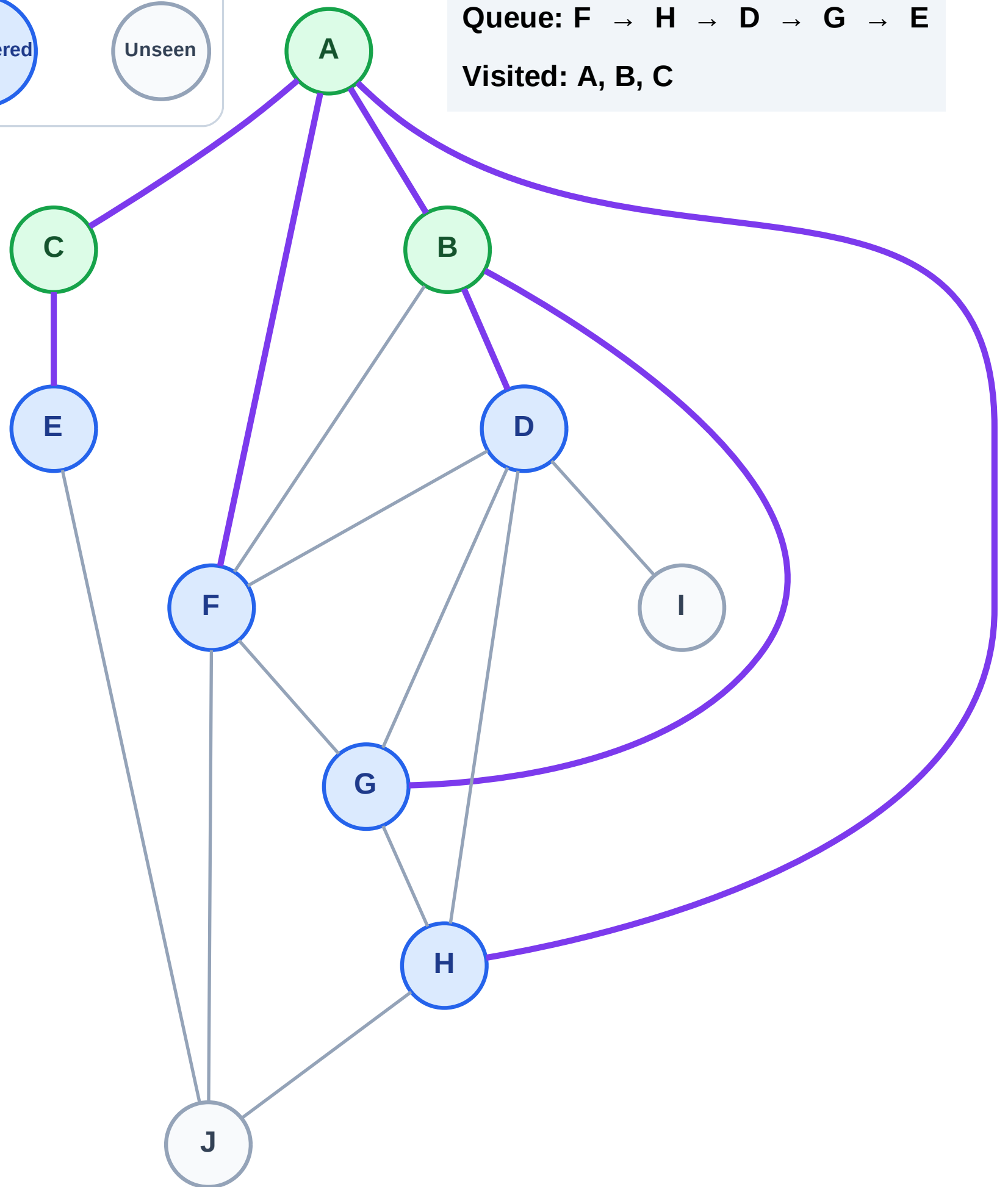


BFS Traversal

Step 7: Discover E from C



Queue: F → H → D → G → E
Visited: A, B, C



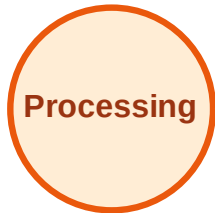
BFS Traversal

Step 8: Process node F

Legend



Complete



Processing



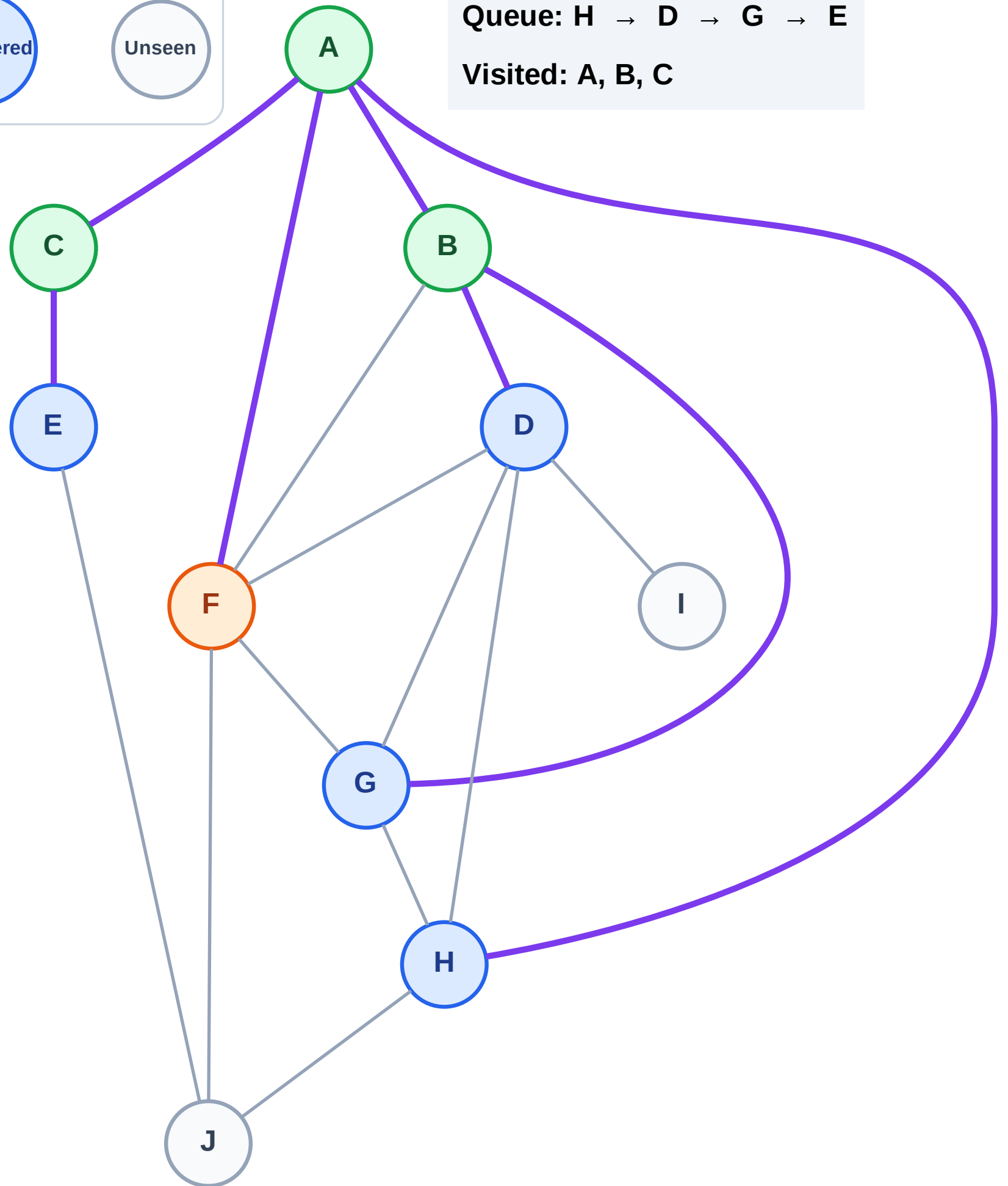
Discovered



Unseen

Queue: H → D → G → E

Visited: A, B, C



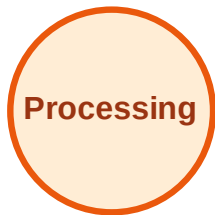
BFS Traversal

Step 9: Discover J from F

Legend



Complete



Processing



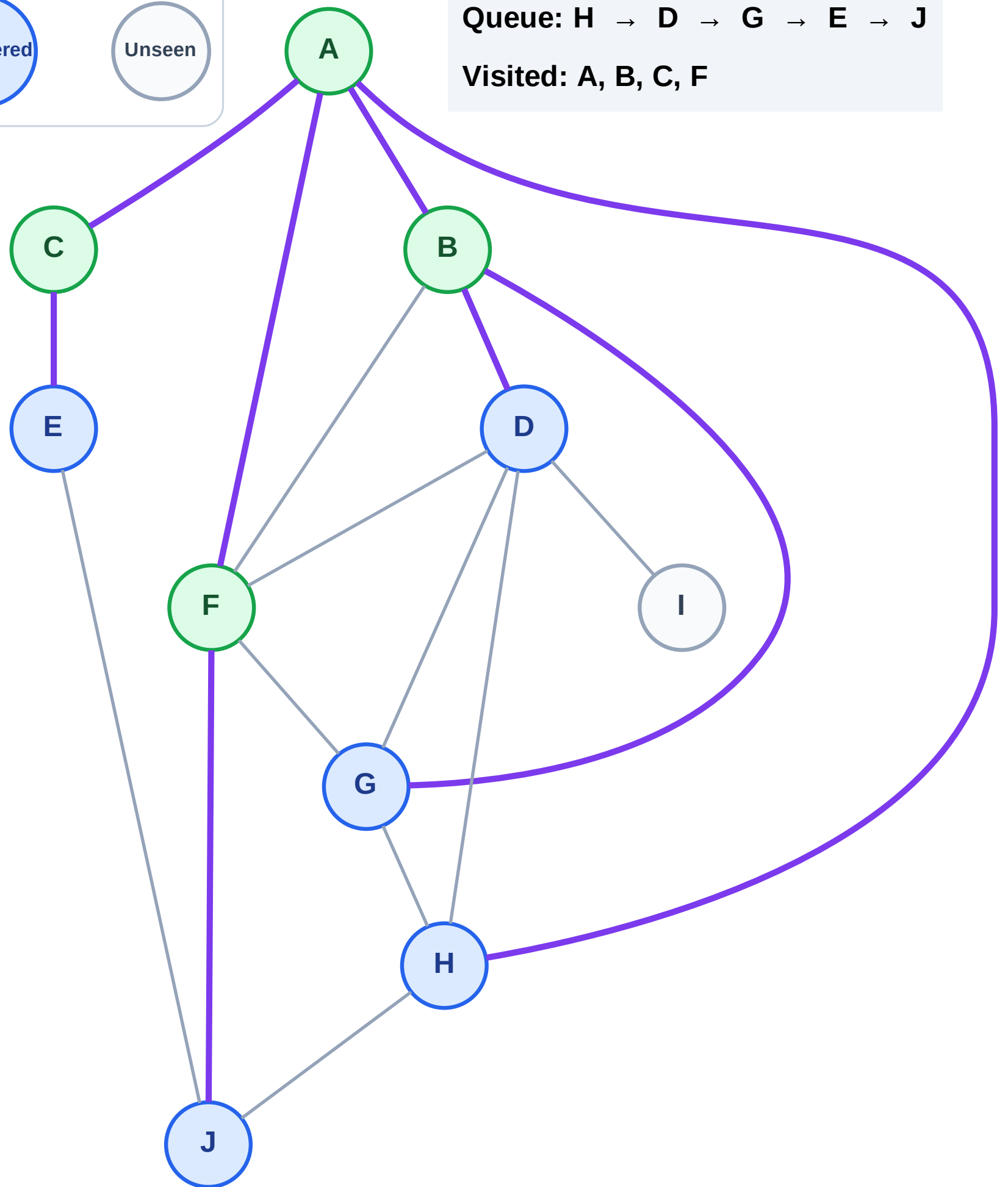
Discovered



Unseen

Queue: H → D → G → E → J

Visited: A, B, C, F



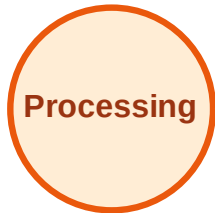
BFS Traversal

Step 10: Process node H

Legend



Complete



Processing



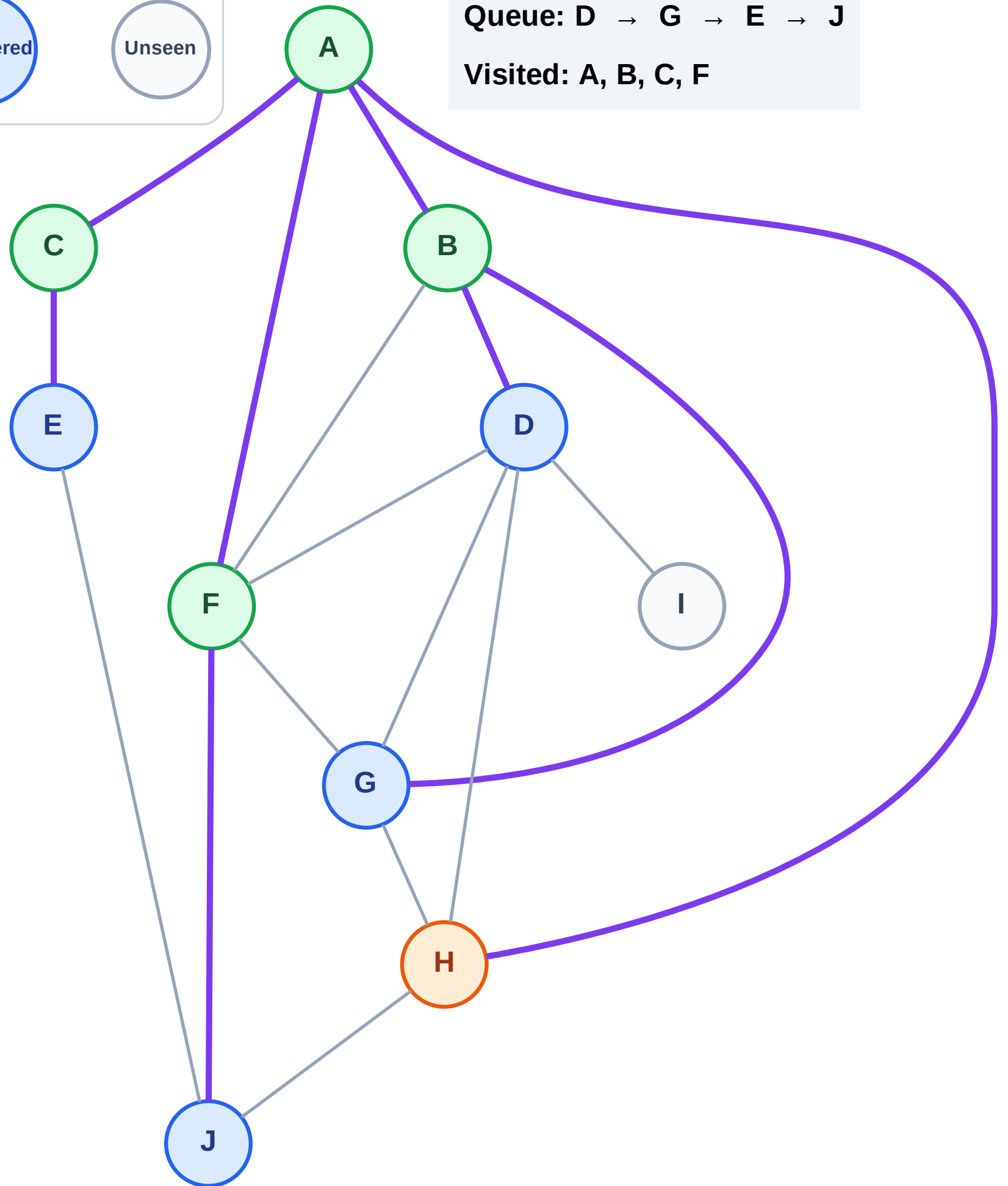
Discovered



Unseen

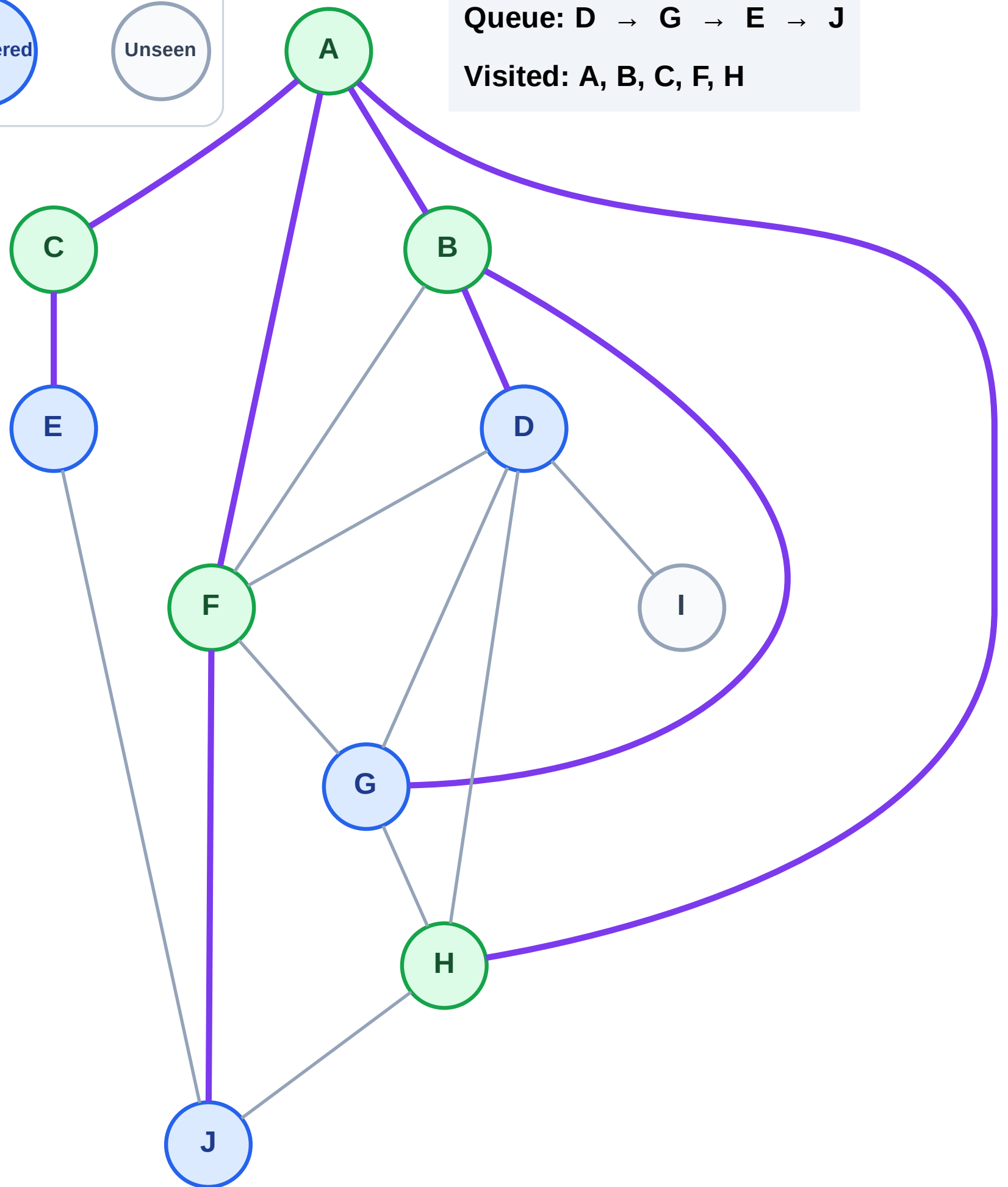
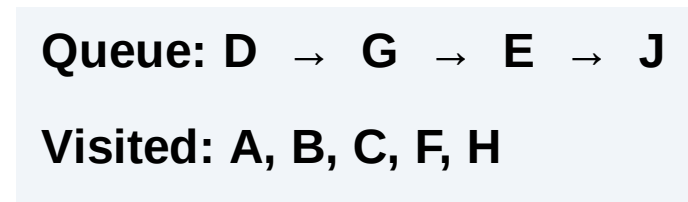
Queue: D → G → E → J

Visited: A, B, C, F



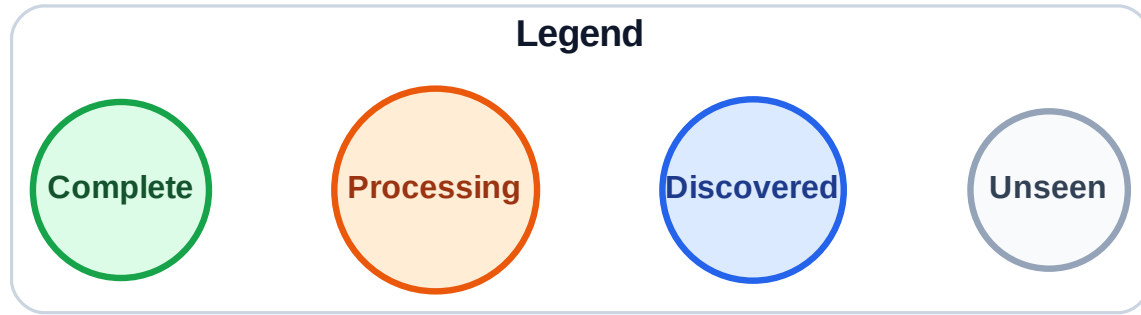
Step 11: No new neighbors from H

Step 11: No new neighbors from H



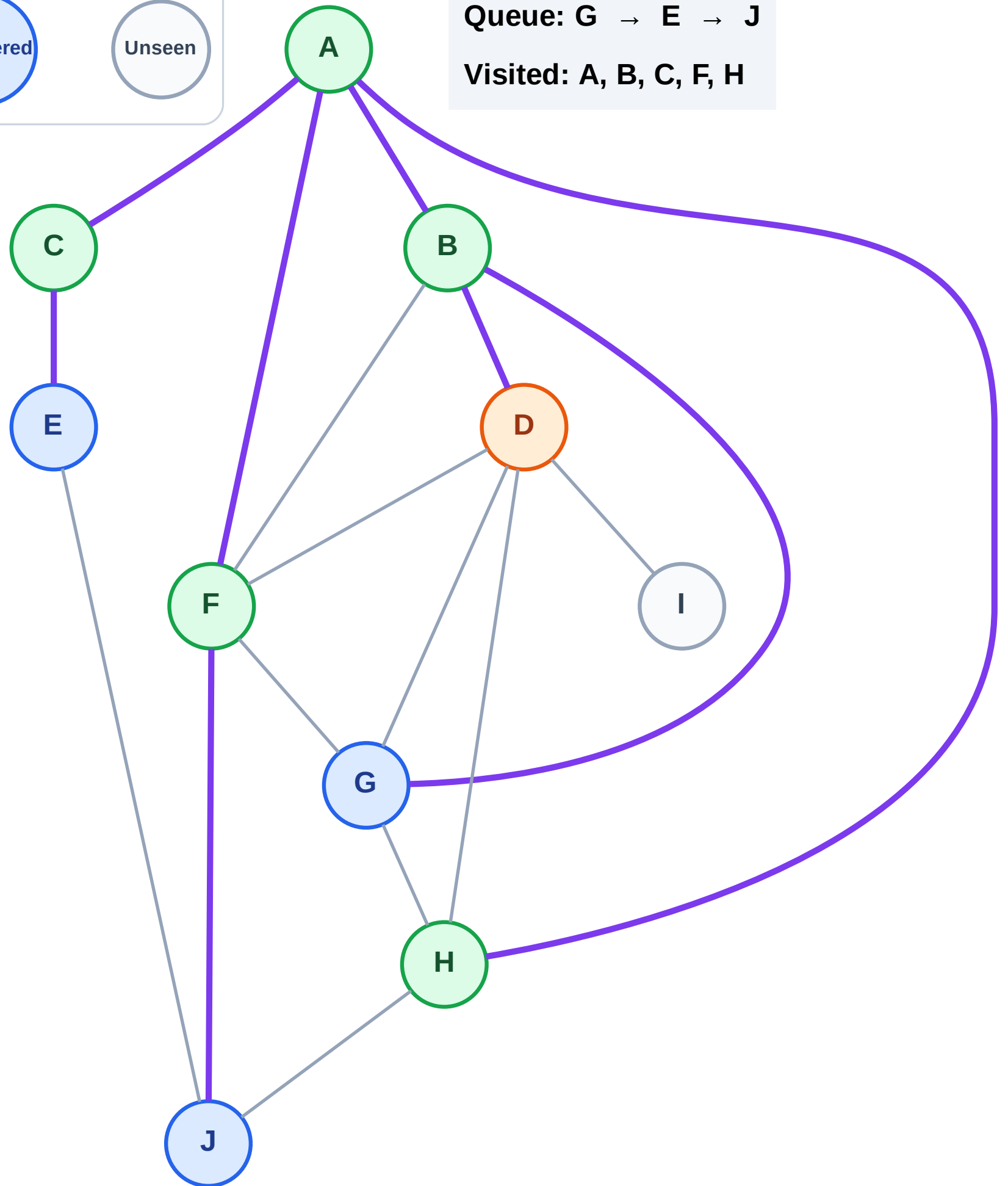
Step 12: Process node D

Step 12: Process node D



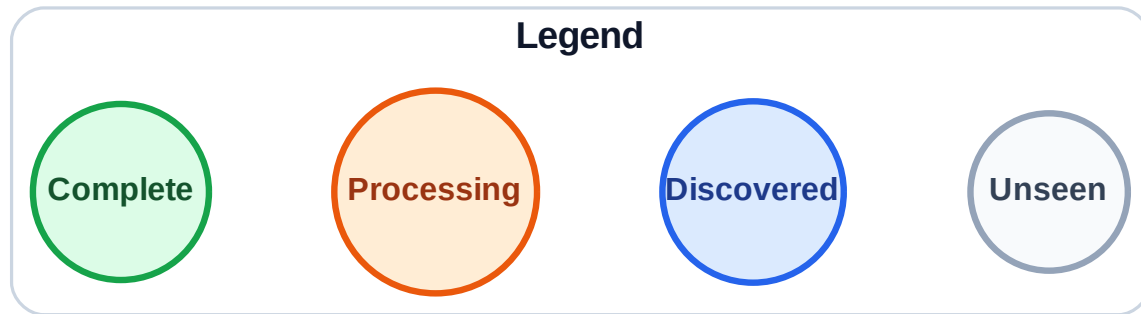
Queue: G → E → J

Visited: A, B, C, F, H

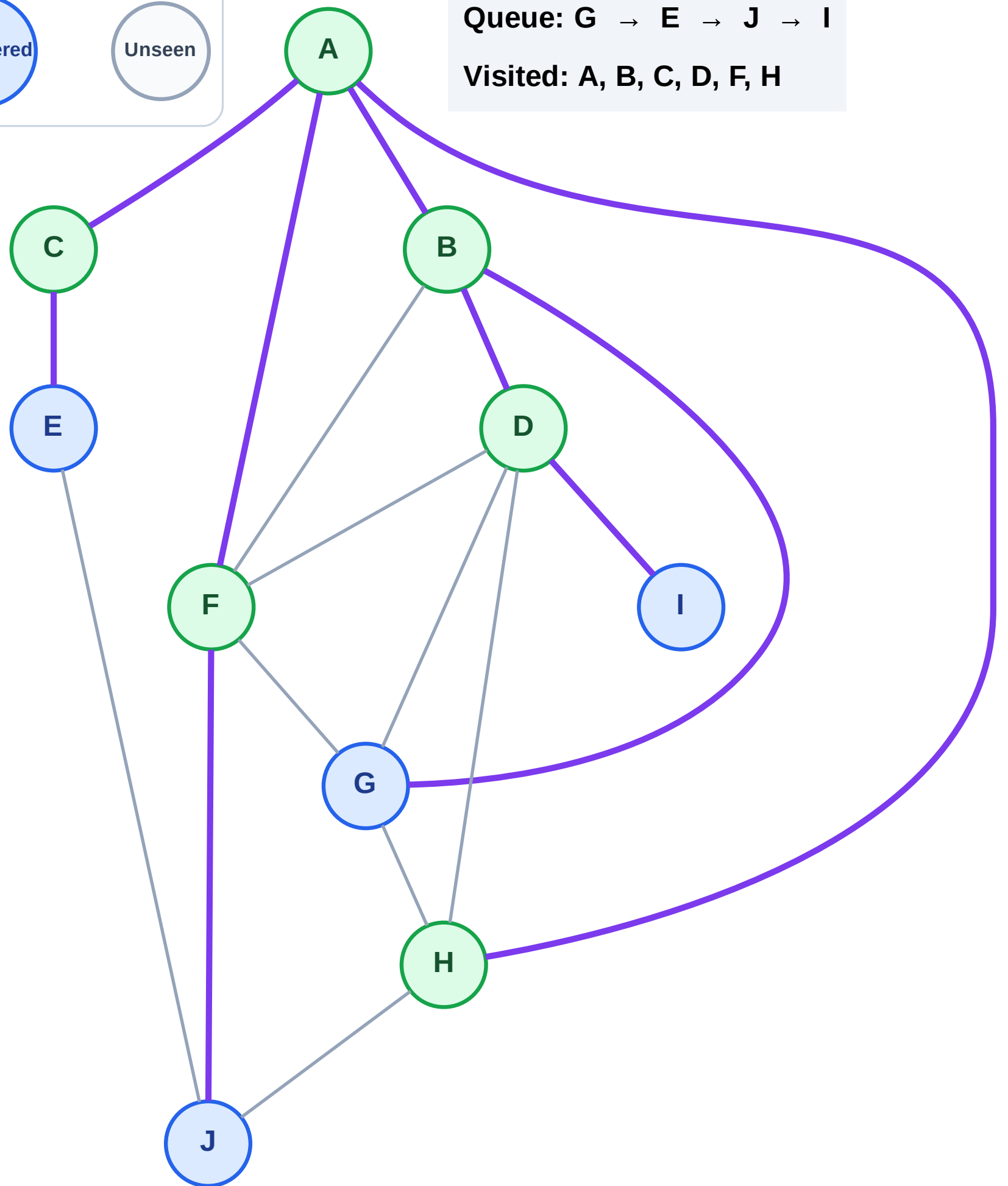


BFS Traversal

Step 13: Discover I from D

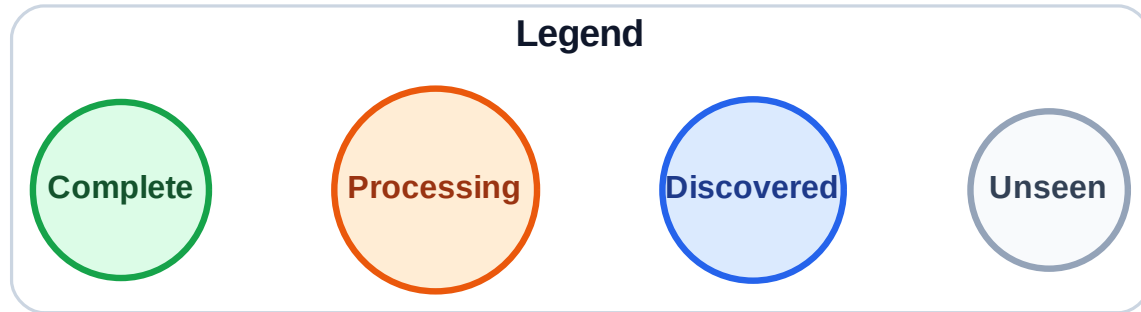


Queue: G → E → J → I
Visited: A, B, C, D, F, H



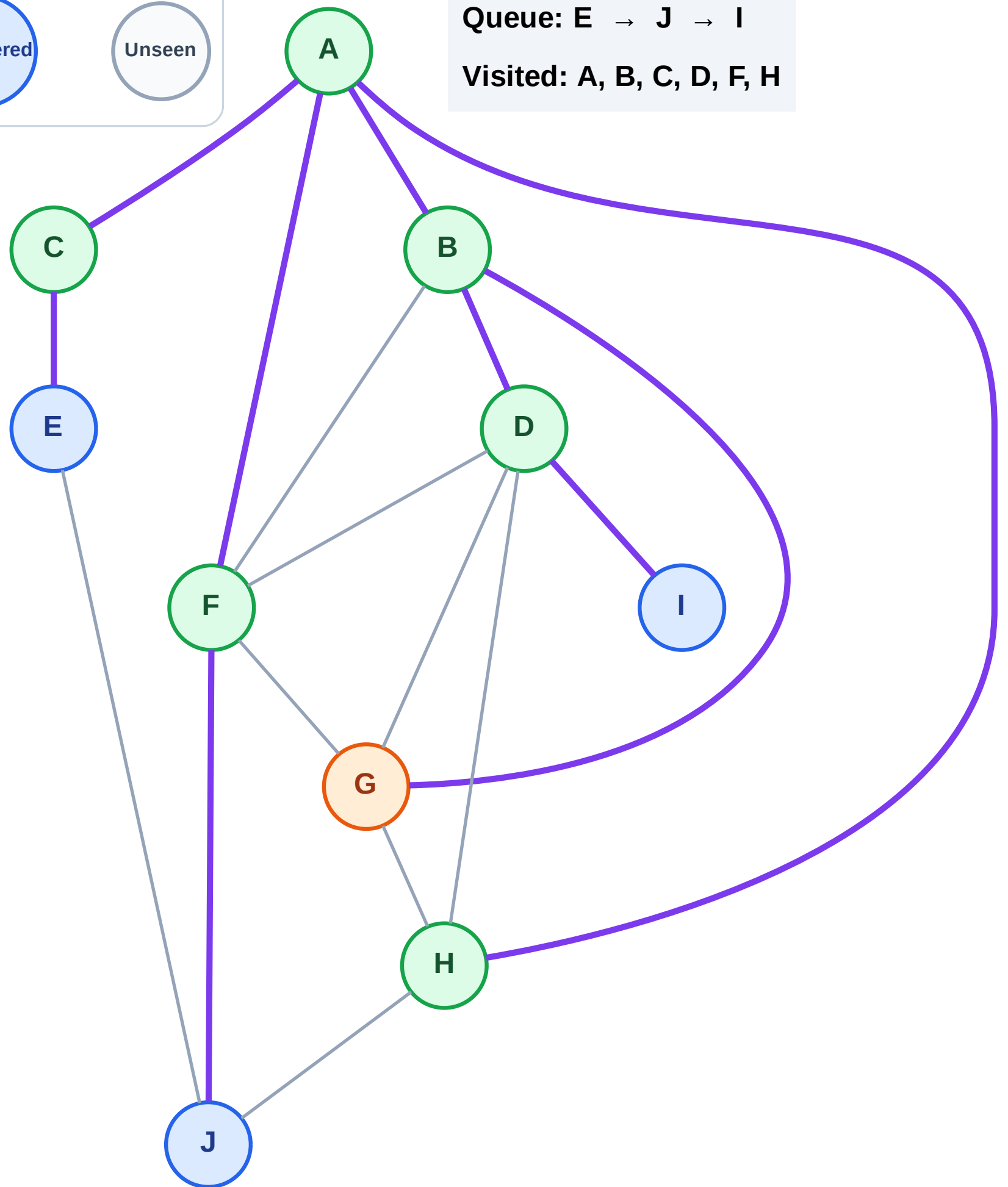
Step 14: Process node G

Step 14: Process node G



Queue: E → J → I

Visited: A, B, C, D, F, H



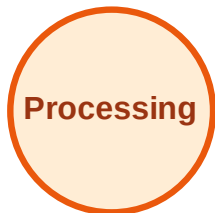
BFS Traversal

Step 15: No new neighbors from G

Legend



Complete



Processing



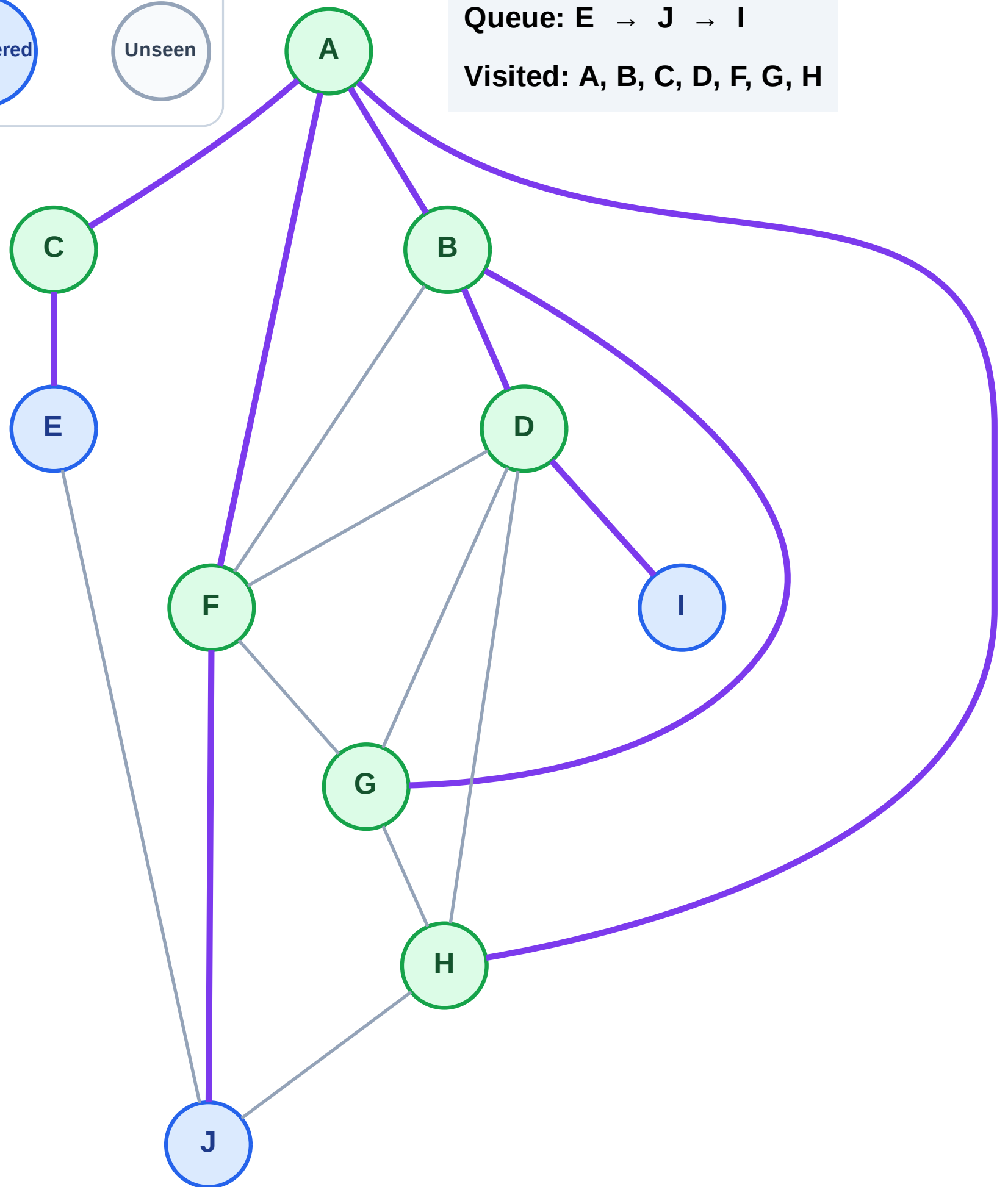
Discovered



Unseen

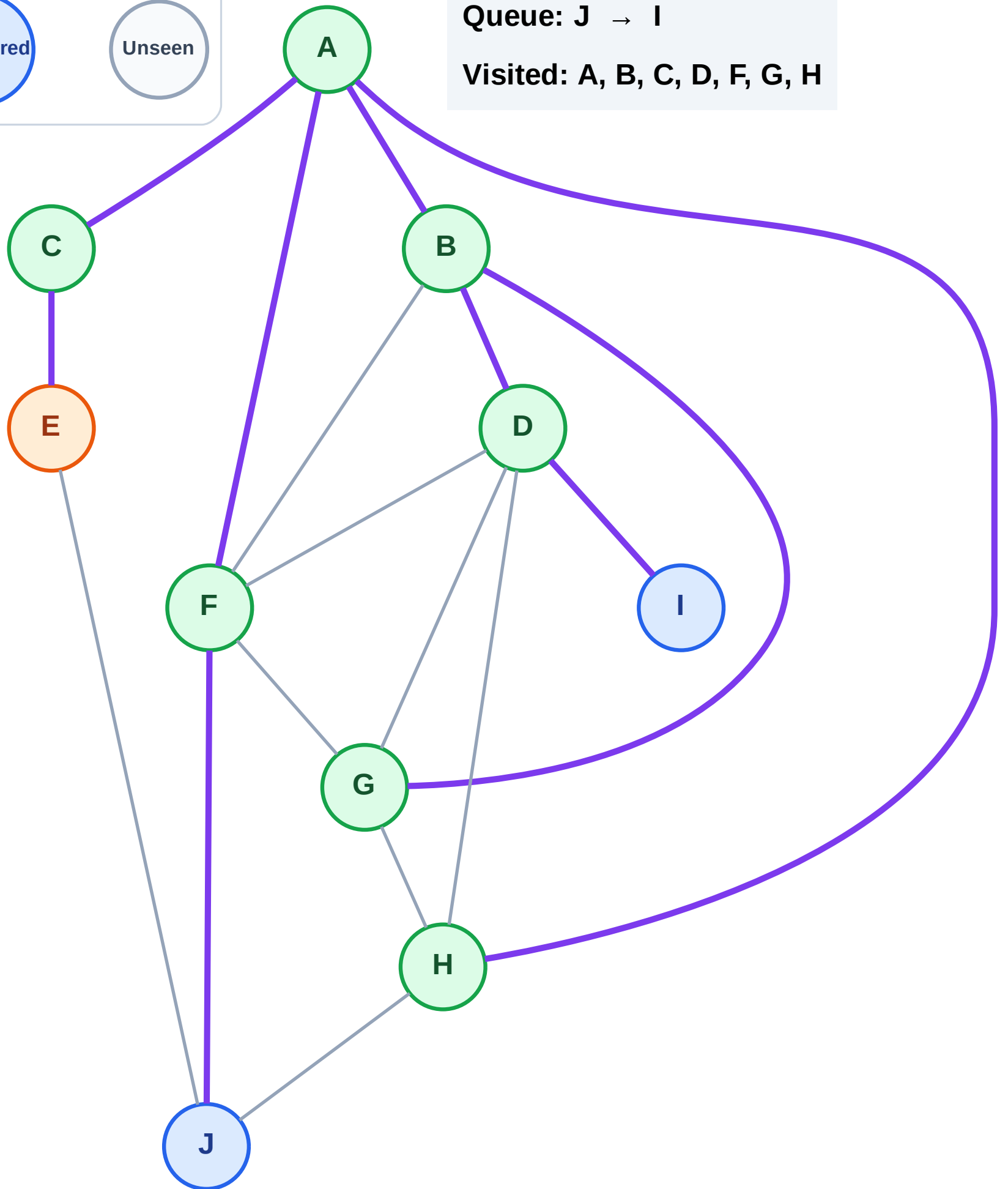
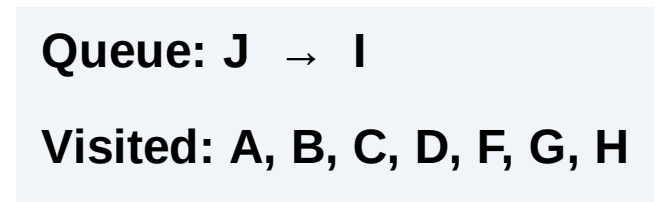
Queue: E → J → I

Visited: A, B, C, D, F, G, H



Step 16: Process node E

Step 16: Process node E



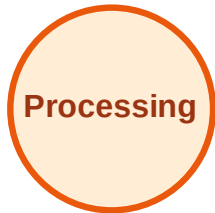
BFS Traversal

Step 17: No new neighbors from E

Legend



Complete



Processing



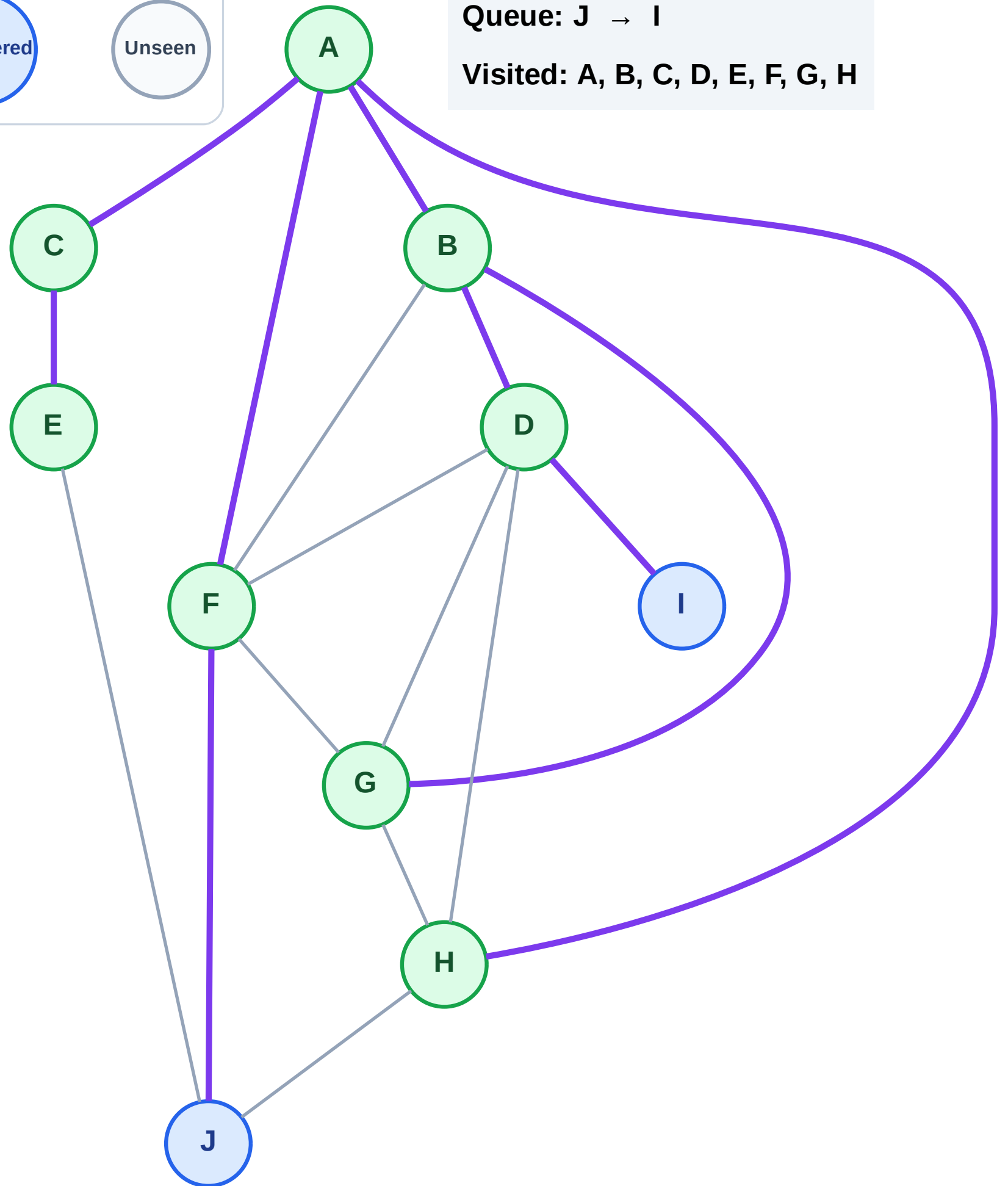
Discovered



Unseen

Queue: J → I

Visited: A, B, C, D, E, F, G, H



BFS Traversal

Step 18: Process node J

Legend

Complete

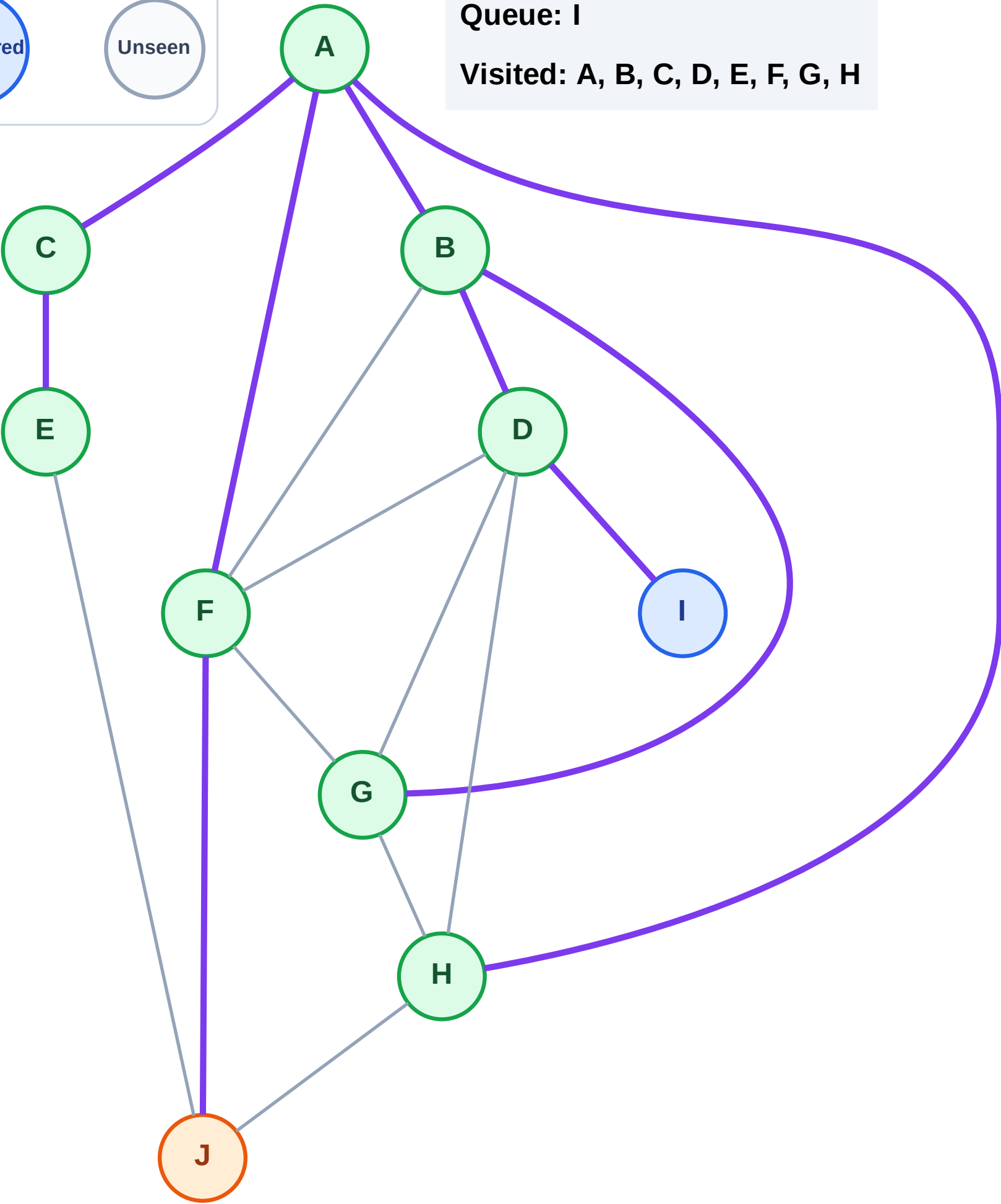
Processing

Discovered

Unseen

Queue: I

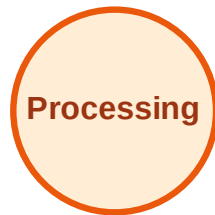
Visited: A, B, C, D, E, F, G, H



BFS Traversal

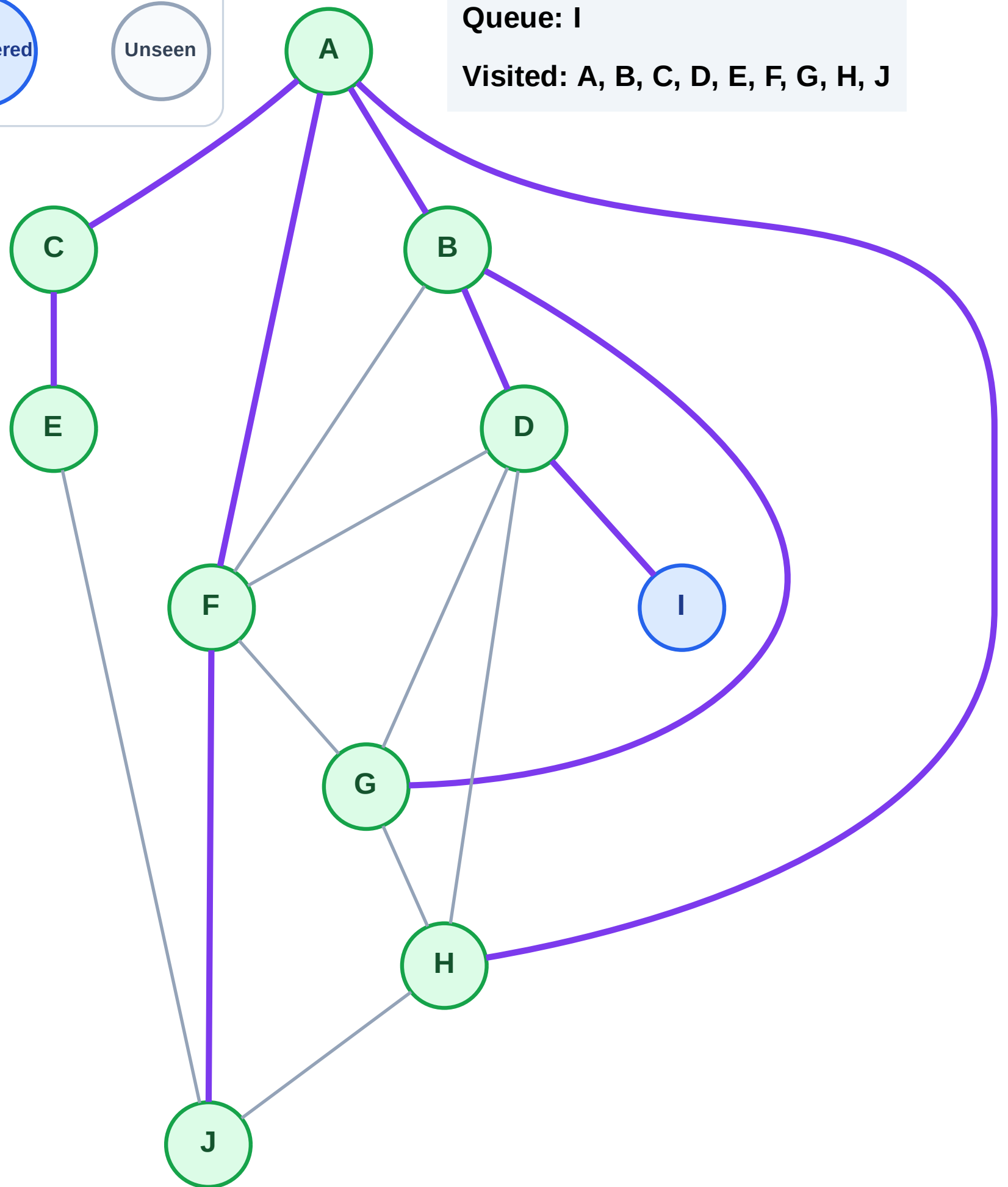
Step 19: No new neighbors from J

Legend

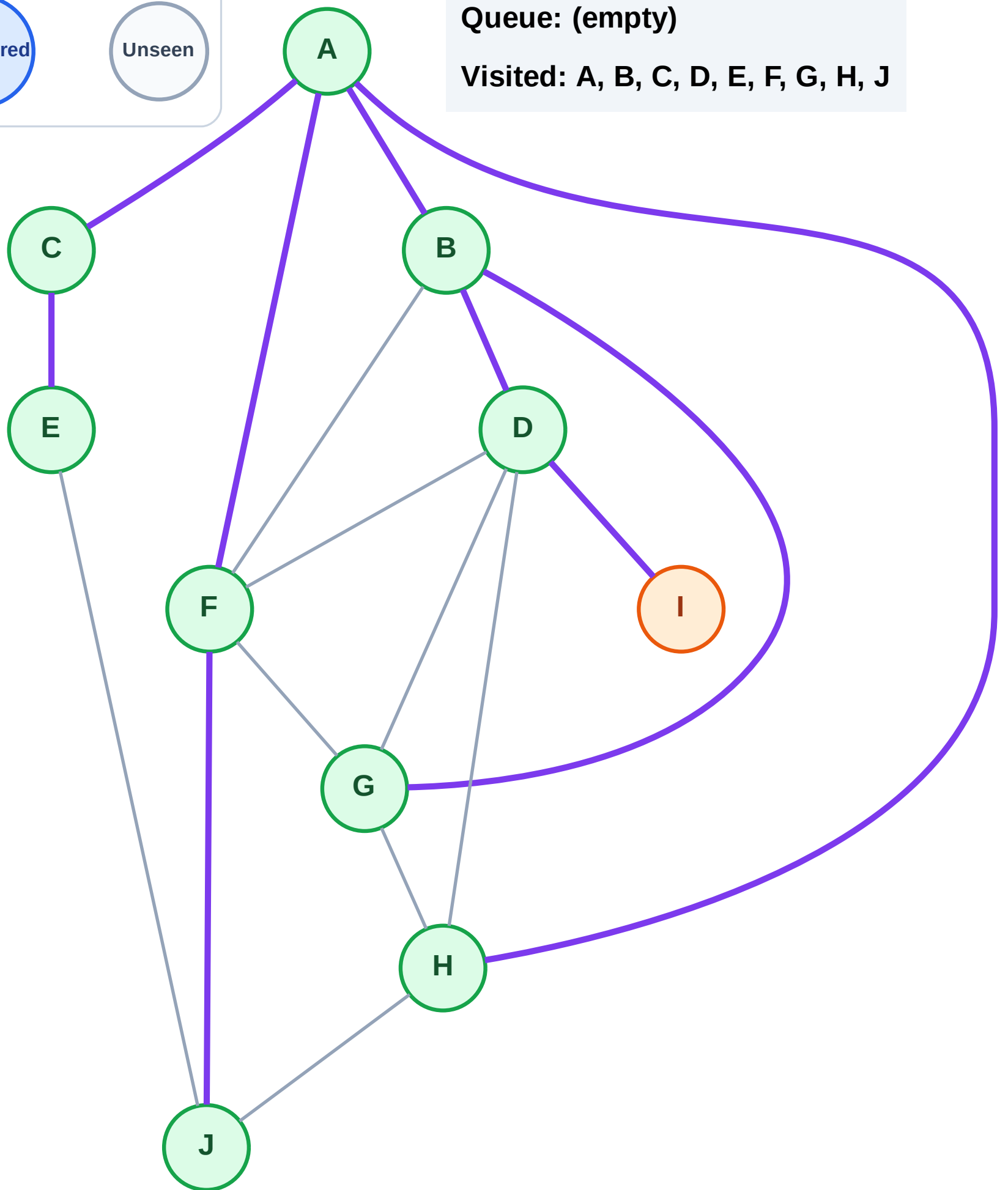
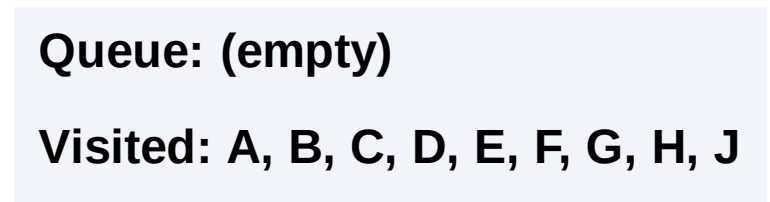


Queue: I

Visited: A, B, C, D, E, F, G, H, J



Step 20: Process node I



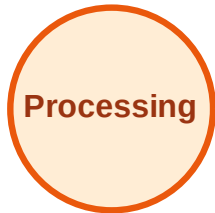
BFS Traversal

Step 21: No new neighbors from I

Legend



Complete



Processing



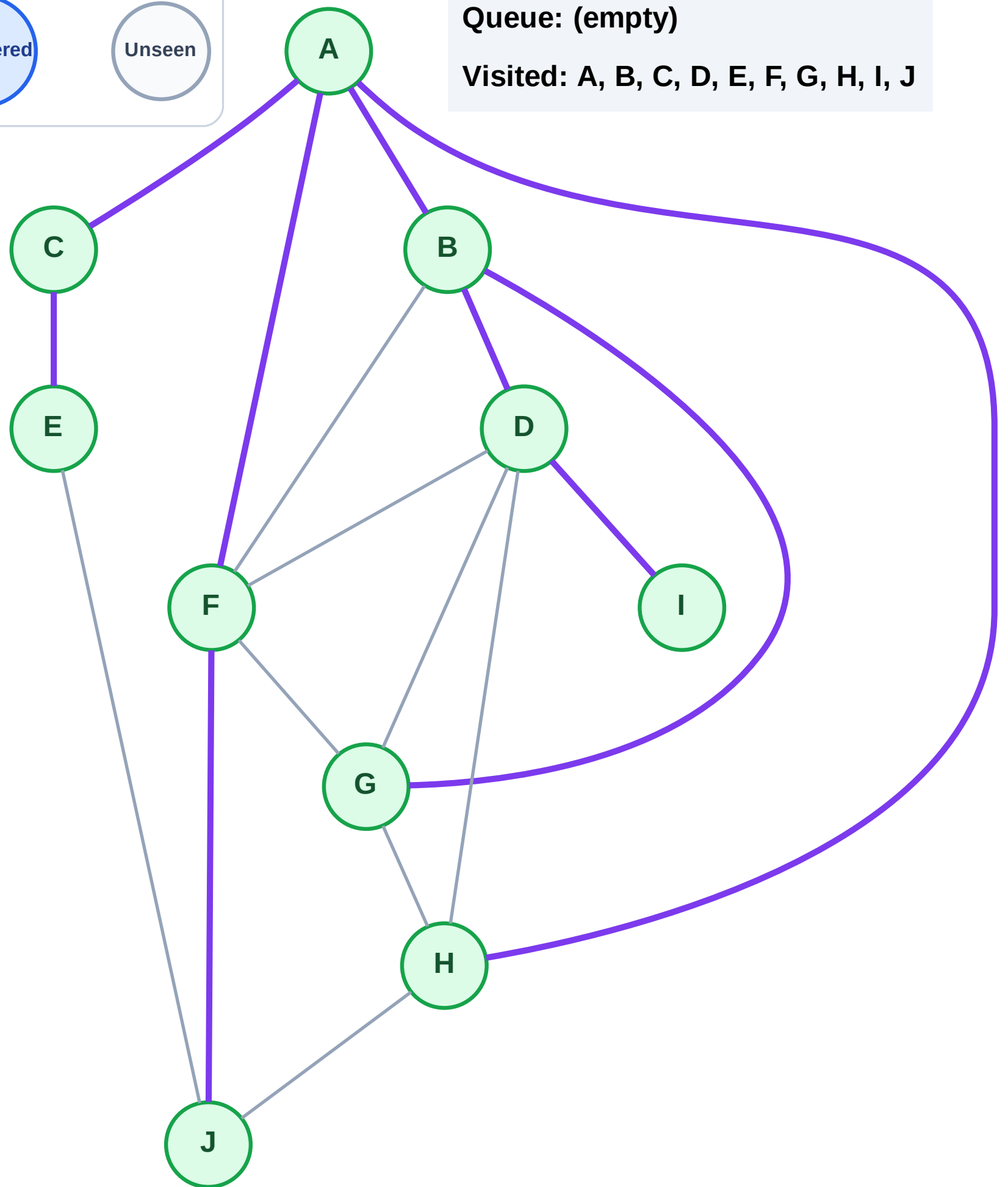
Discovered



Unseen

Queue: (empty)

Visited: A, B, C, D, E, F, G, H, I, J



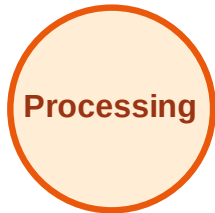
DFS Traversal

Step 1: Start at A

Legend



Complete



Processing



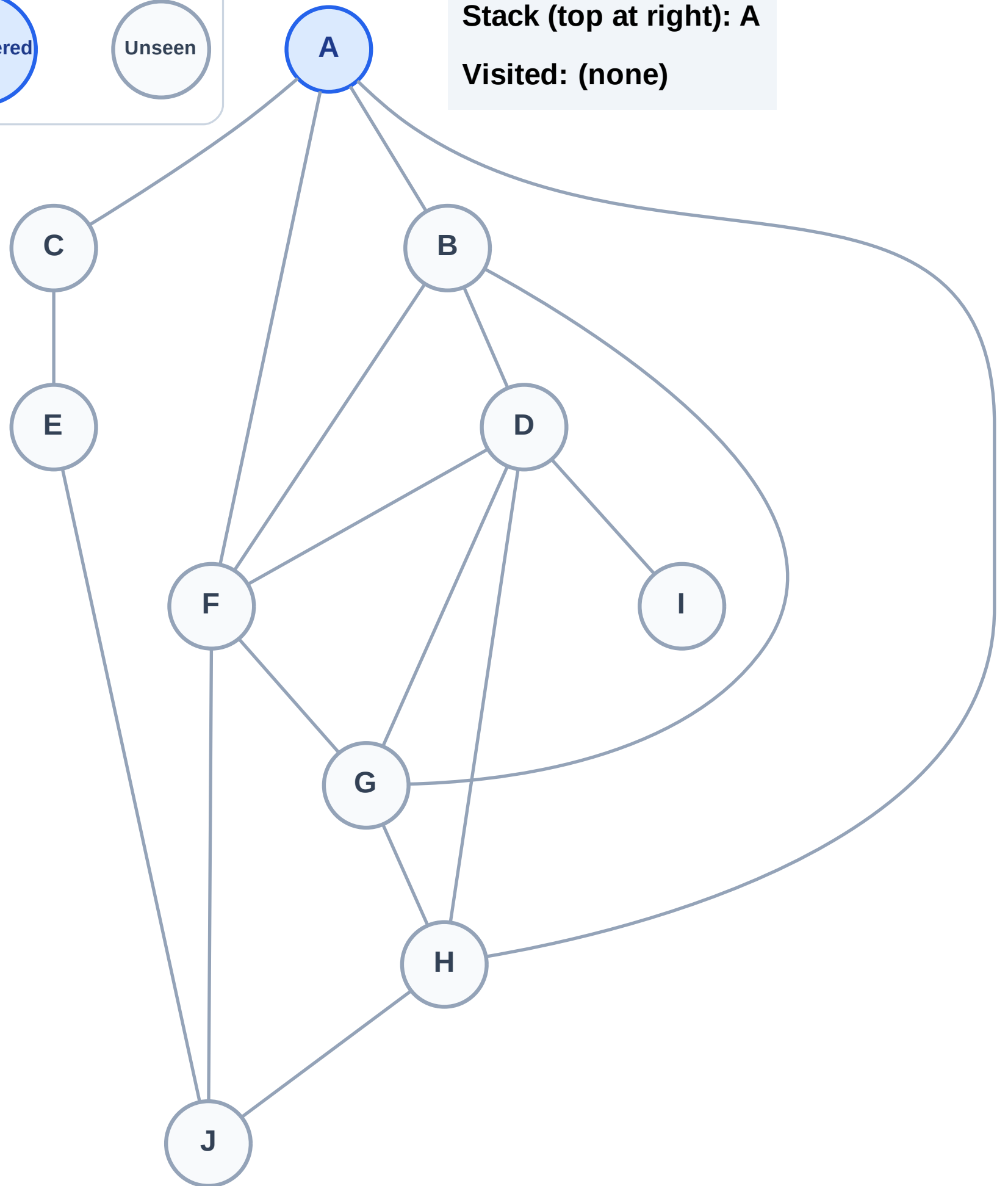
Discovered



Unseen

Stack (top at right): A

Visited: (none)



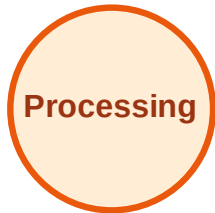
DFS Traversal

Step 2: Process node A

Legend



Complete



Processing



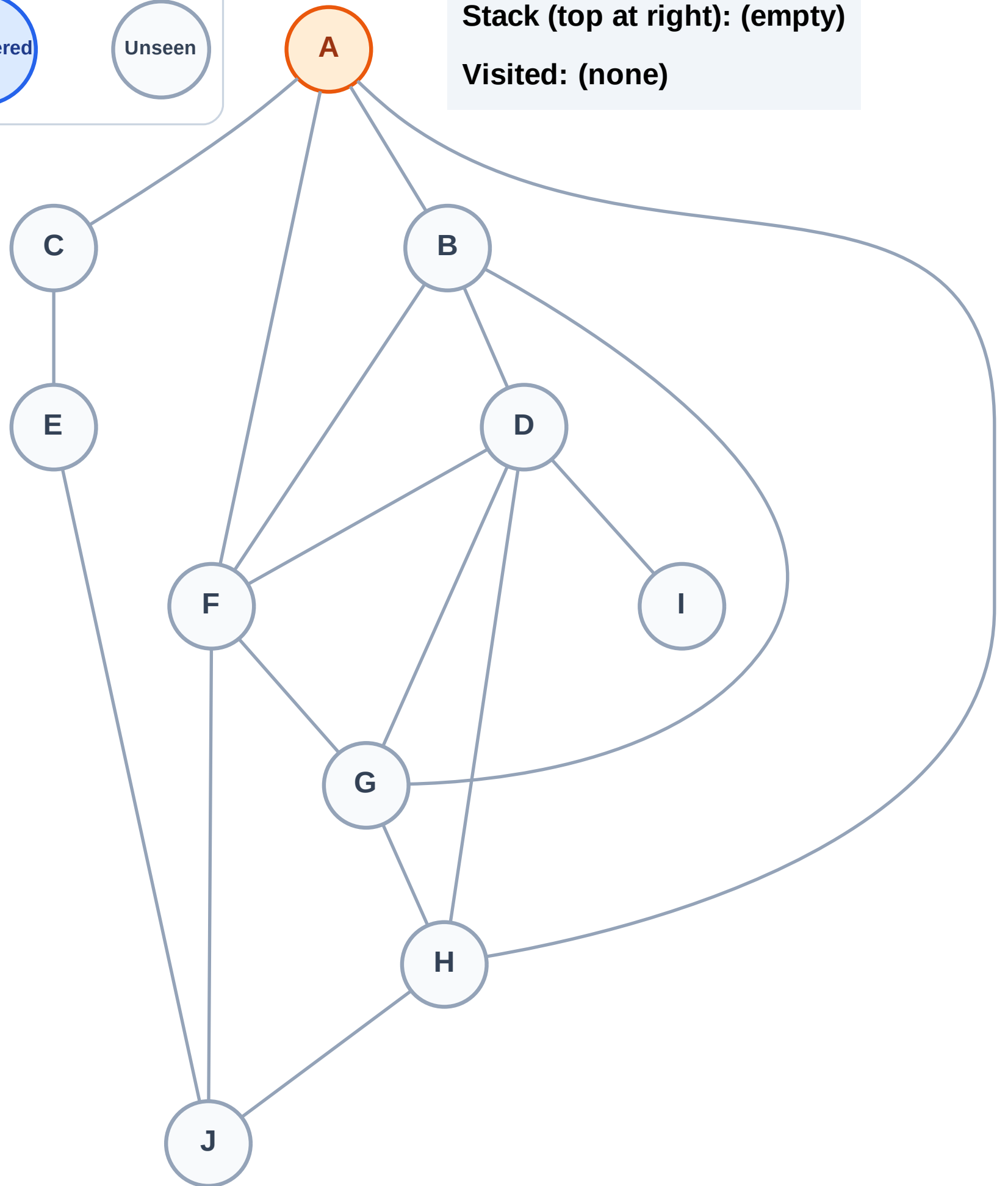
Discovered



Unseen

Stack (top at right): (empty)

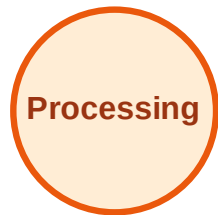
Visited: (none)



DFS Traversal

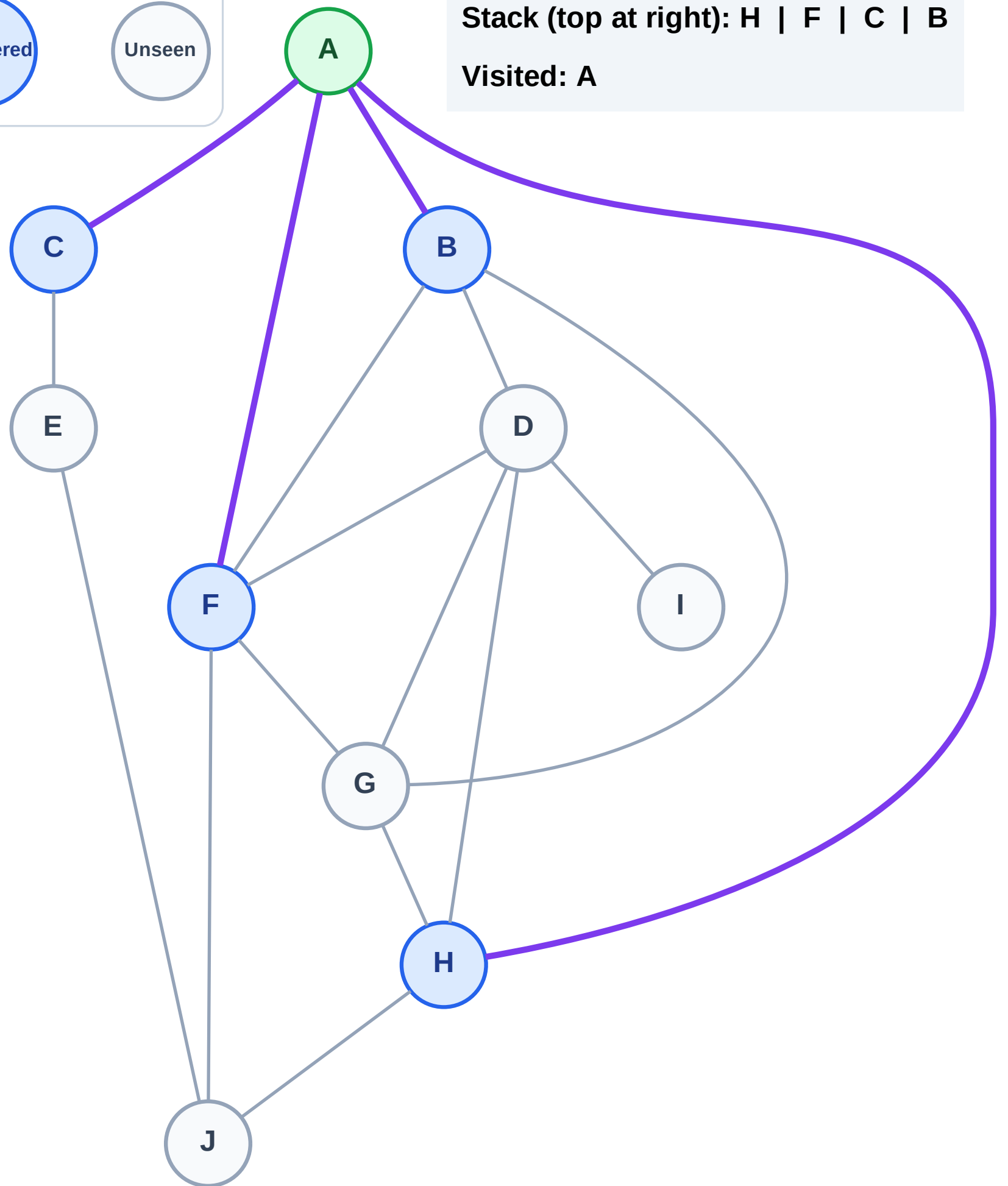
Step 3: Discover H, F, C, B from A

Legend



Stack (top at right): H | F | C | B

Visited: A



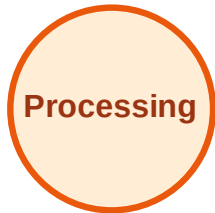
DFS Traversal

Step 4: Process node B

Legend



Complete



Processing



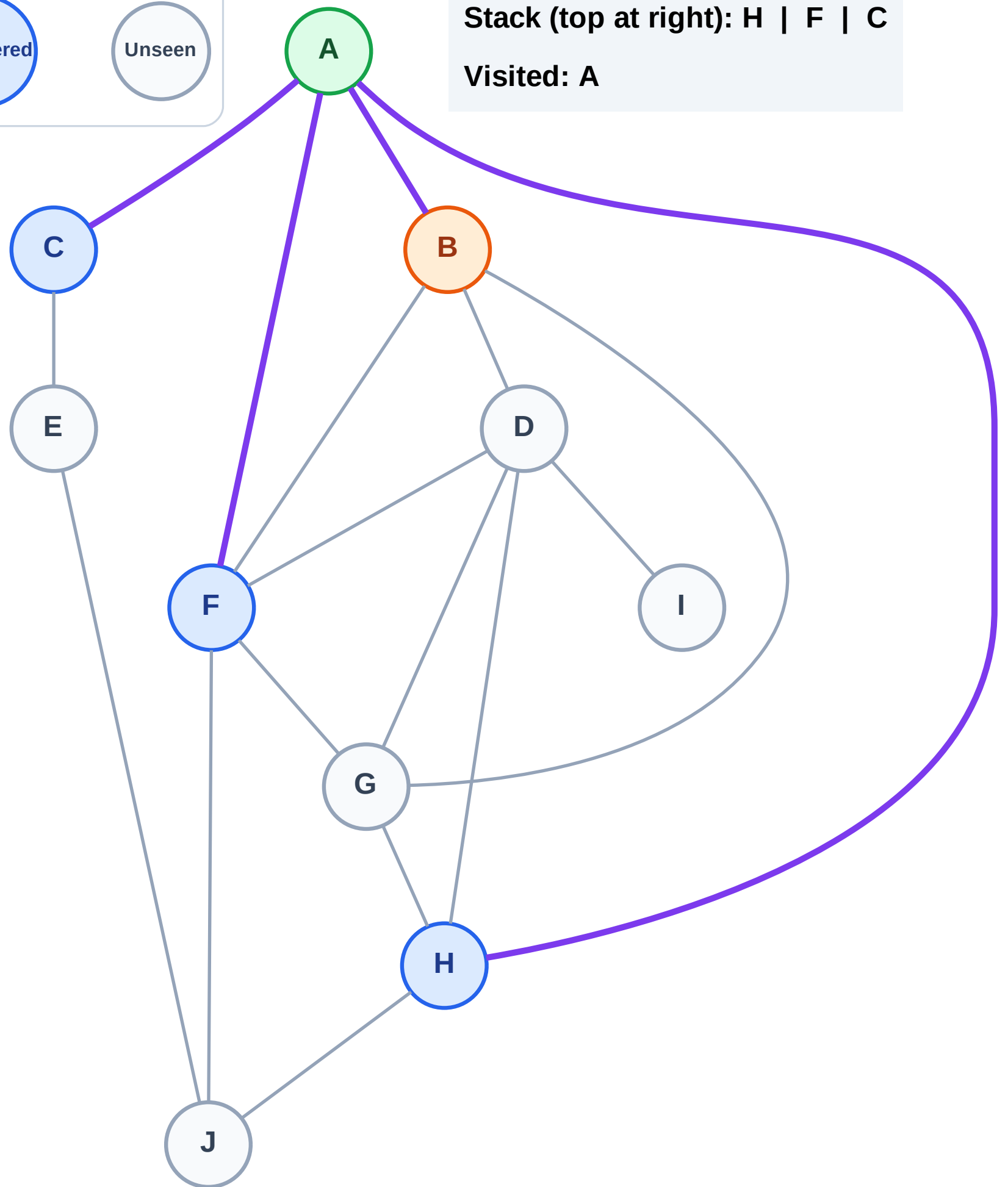
Discovered



Unseen

Stack (top at right): H | F | C

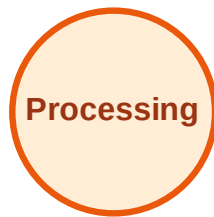
Visited: A



DFS Traversal

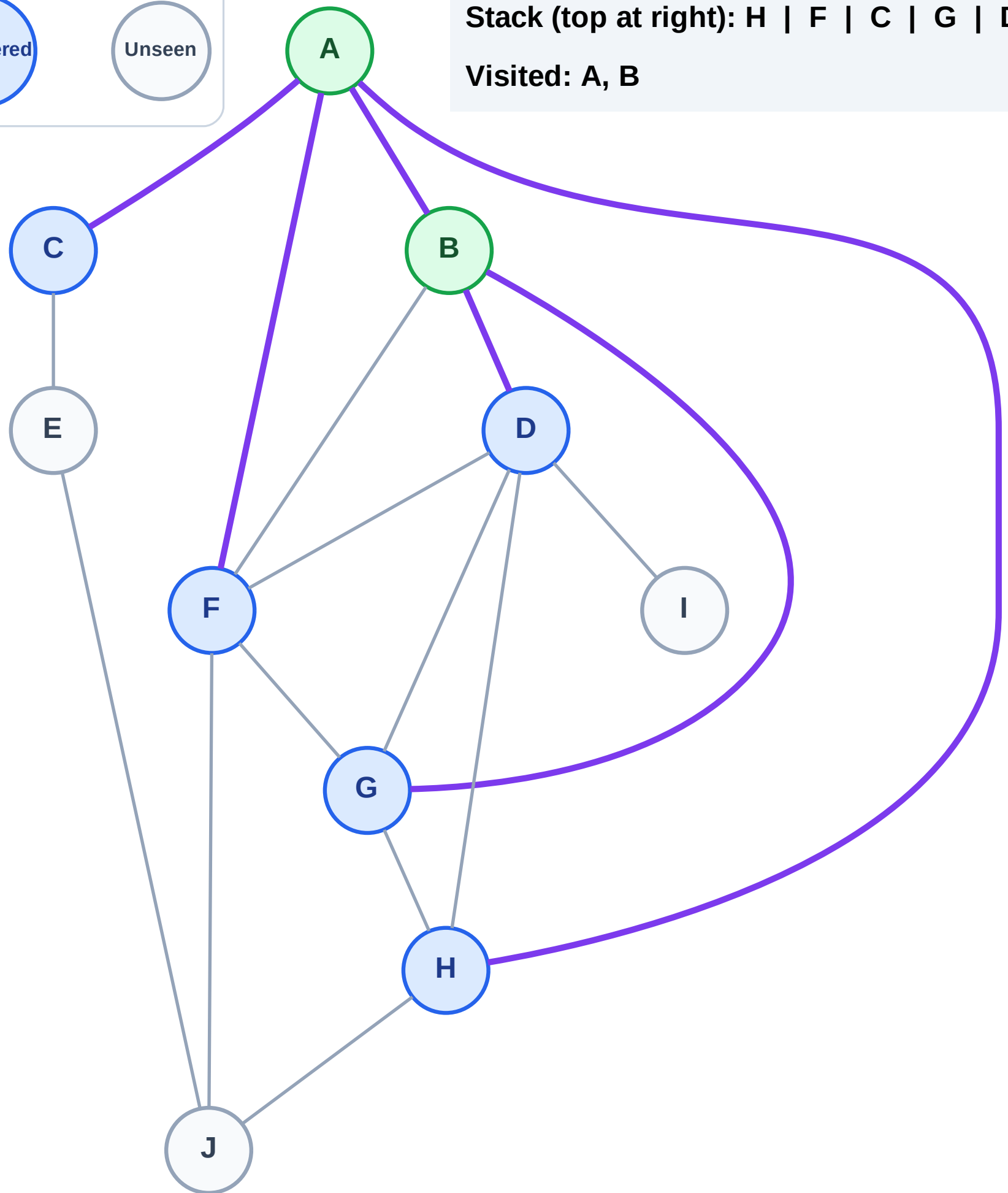
Step 5: Discover G, D from B

Legend



Stack (top at right): H | F | C | G | D

Visited: A, B



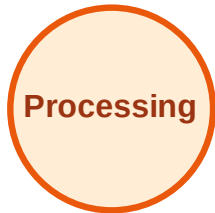
DFS Traversal

Step 6: Process node D

Legend



Complete



Processing



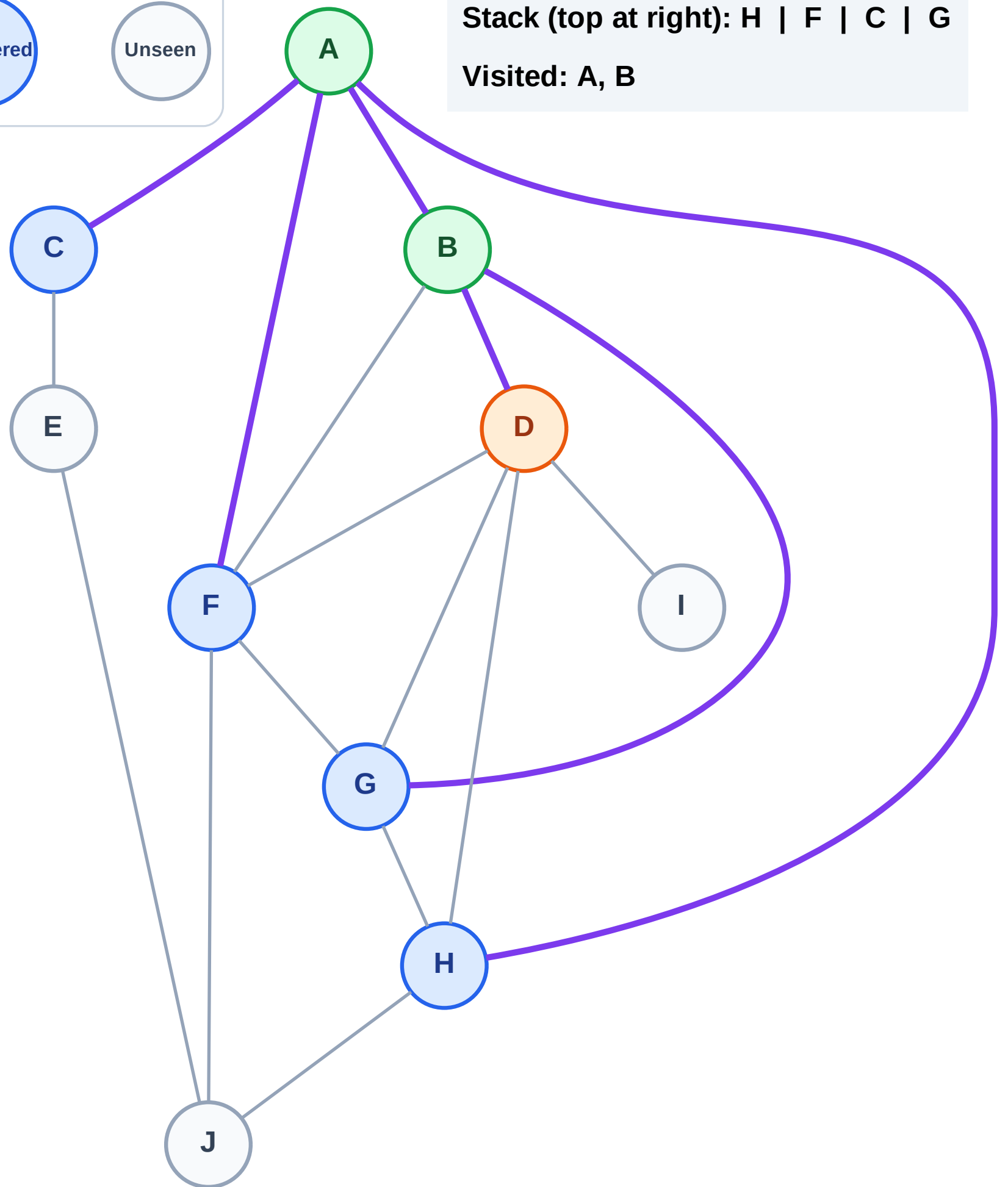
Discovered



Unseen

Stack (top at right): H | F | C | G

Visited: A, B



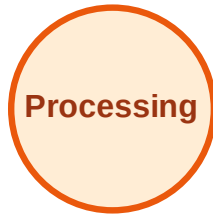
DFS Traversal

Step 7: Discover I from D

Legend



Complete



Processing



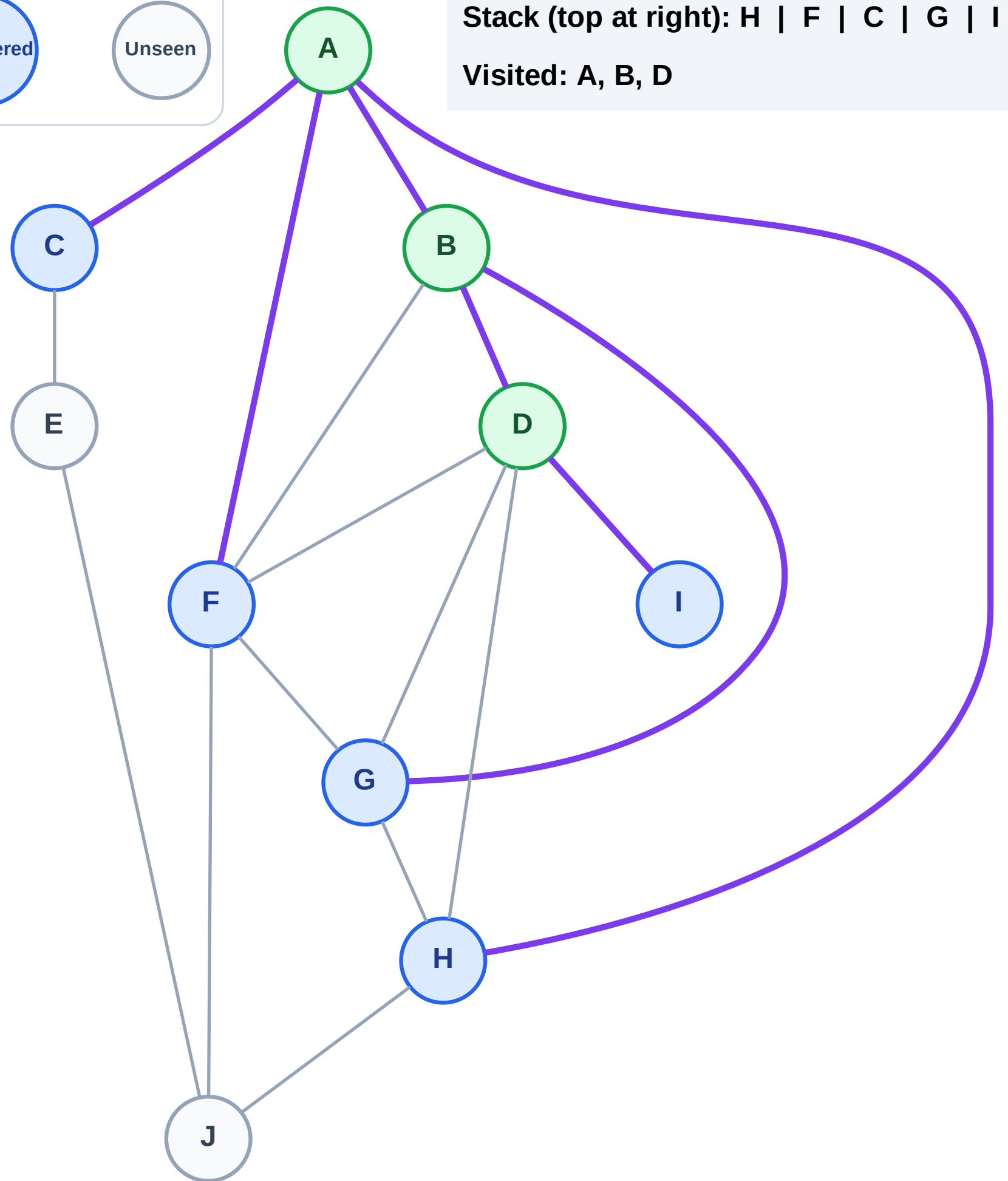
Discovered



Unseen

Stack (top at right): H | F | C | G | I

Visited: A, B, D



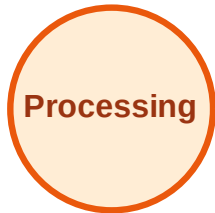
DFS Traversal

Step 8: Process node I

Legend



Complete



Processing



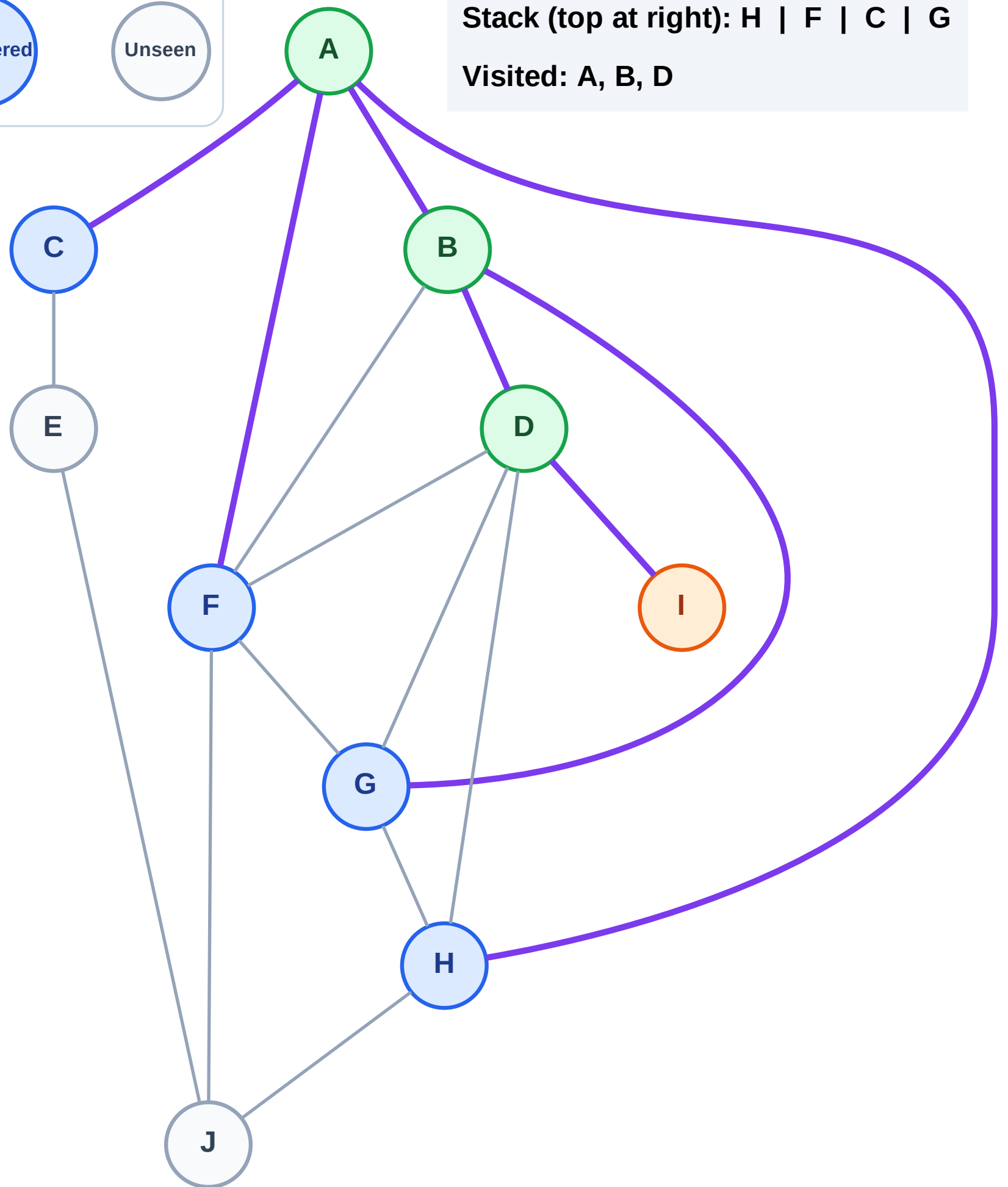
Discovered



Unseen

Stack (top at right): H | F | C | G

Visited: A, B, D

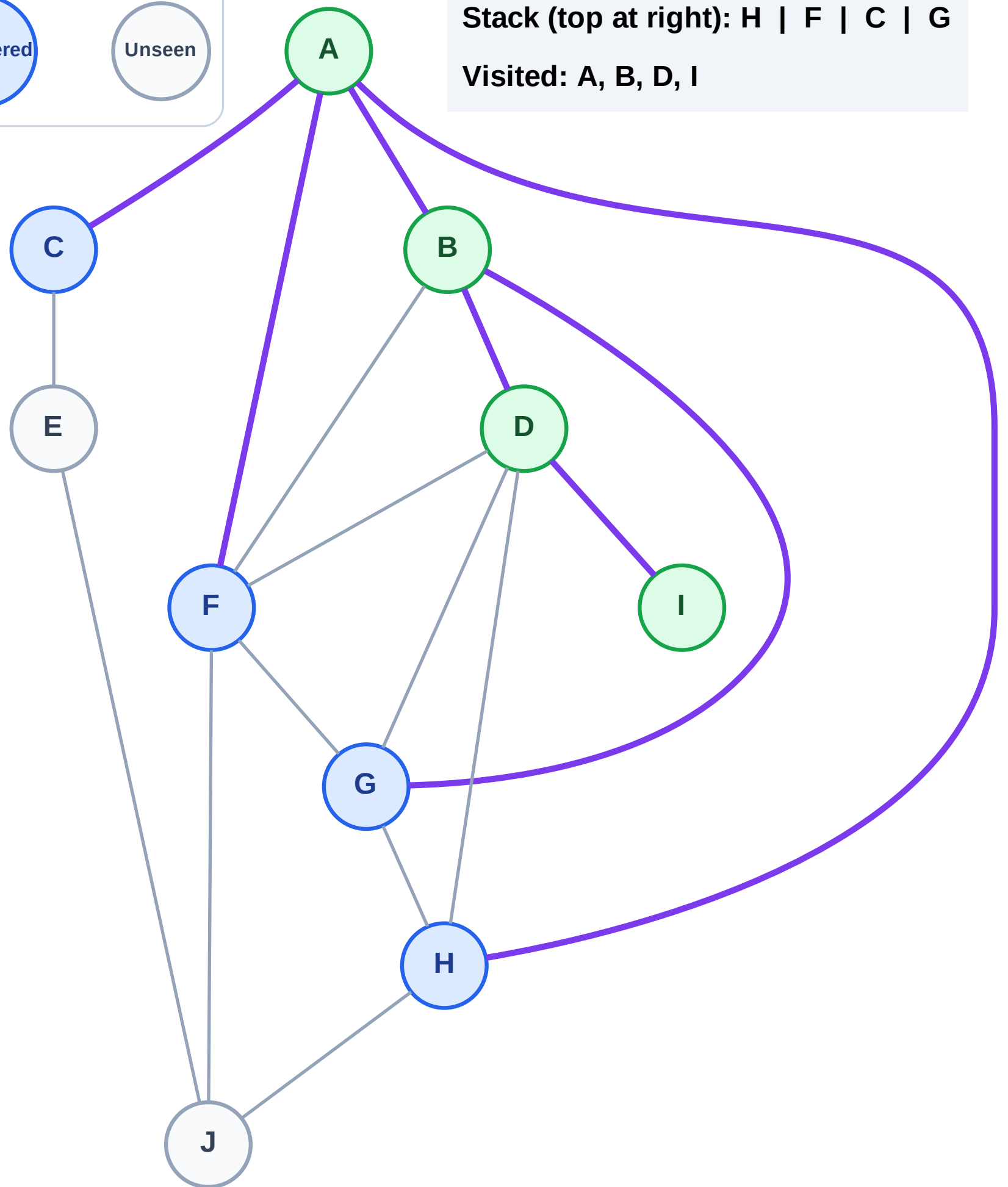


Step 9: No new neighbors from I

Legend



Visited: A, B, D, I



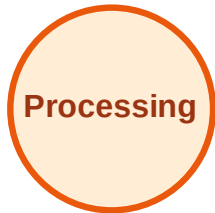
DFS Traversal

Step 10: Process node G

Legend



Complete



Processing



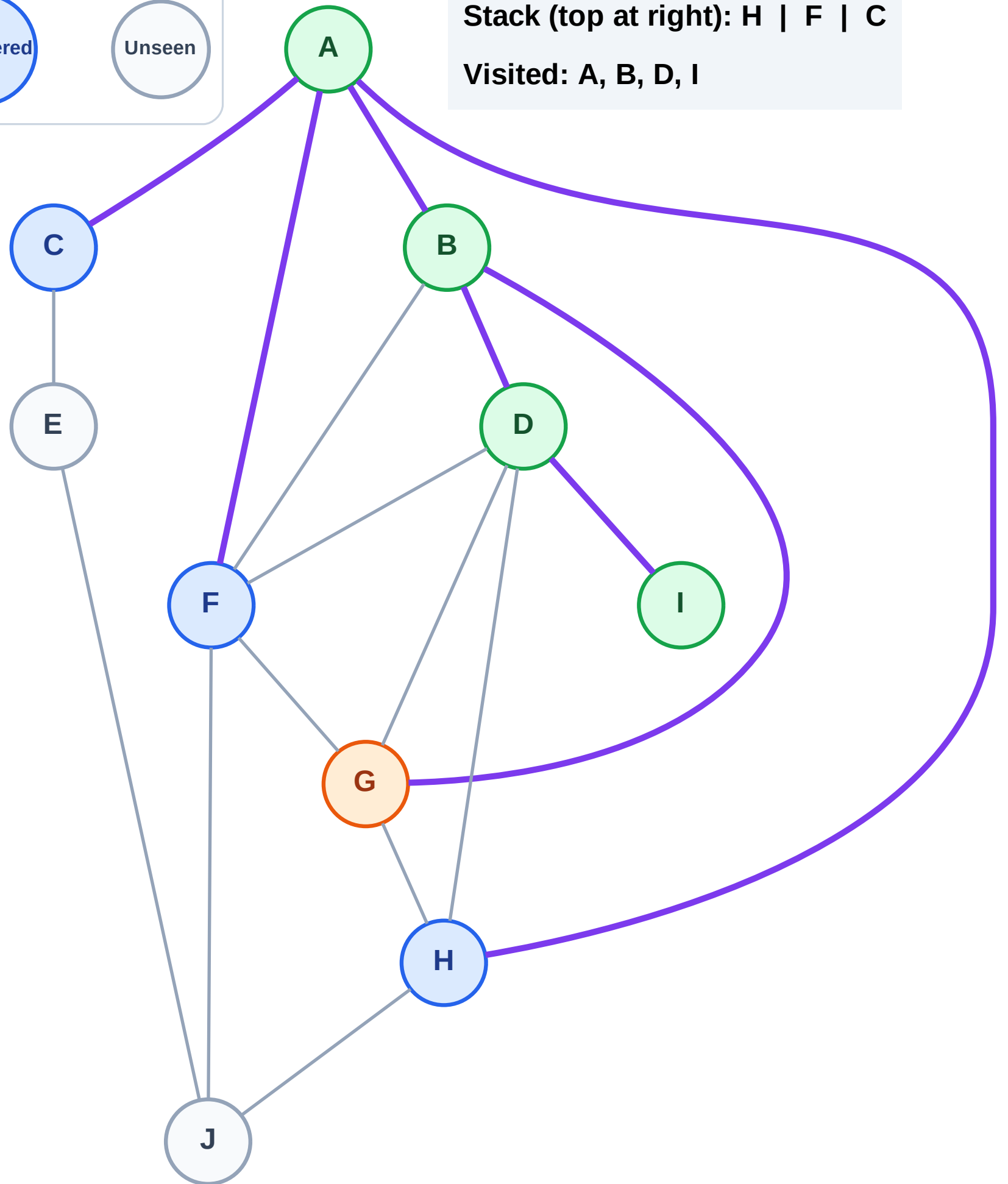
Discovered



Unseen

Stack (top at right): H | F | C

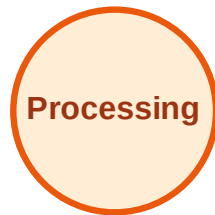
Visited: A, B, D, I



DFS Traversal

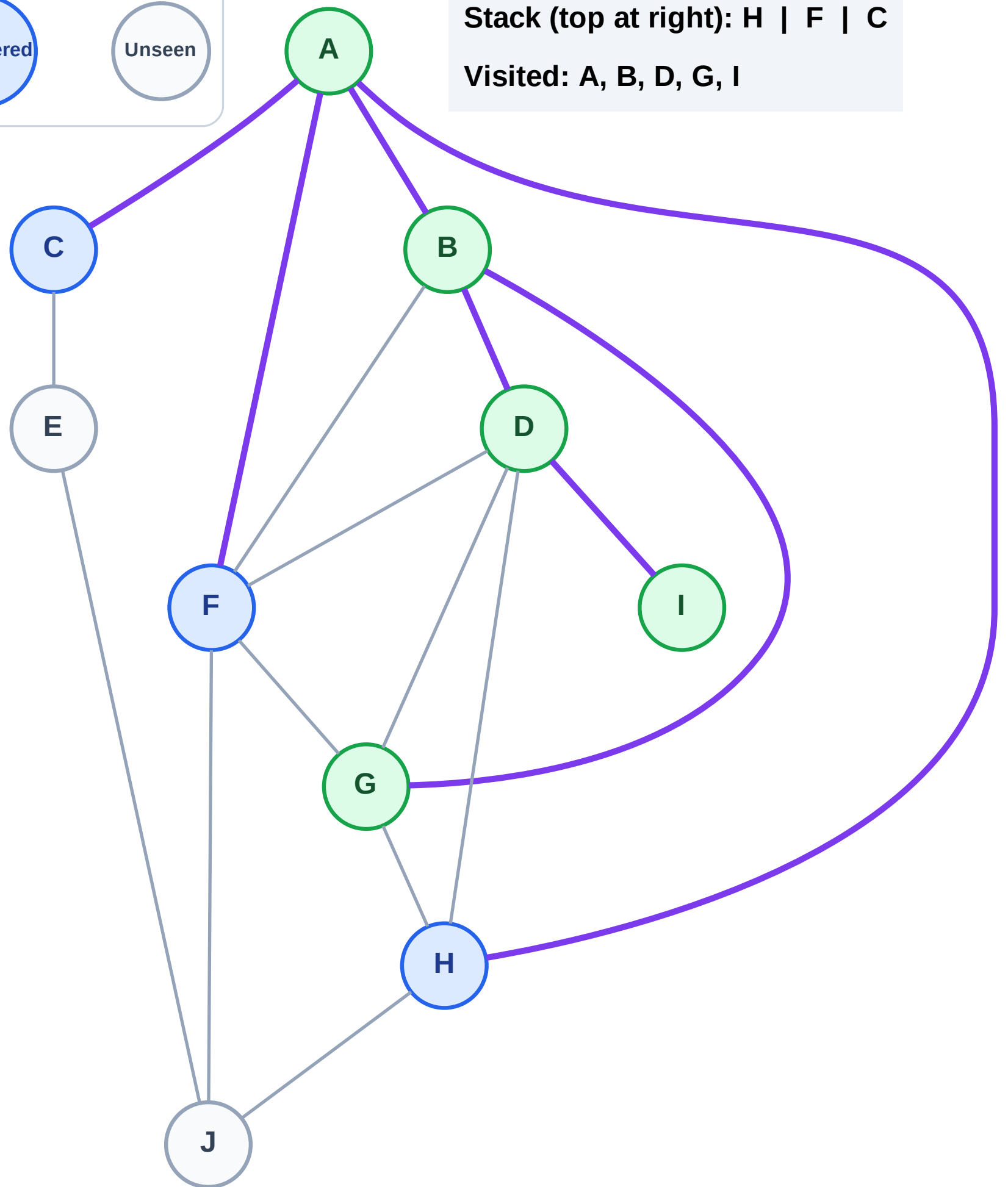
Step 11: No new neighbors from G

Legend



Stack (top at right): H | F | C

Visited: A, B, D, G, I



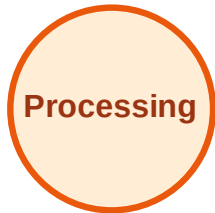
DFS Traversal

Step 12: Process node C

Legend



Complete



Processing



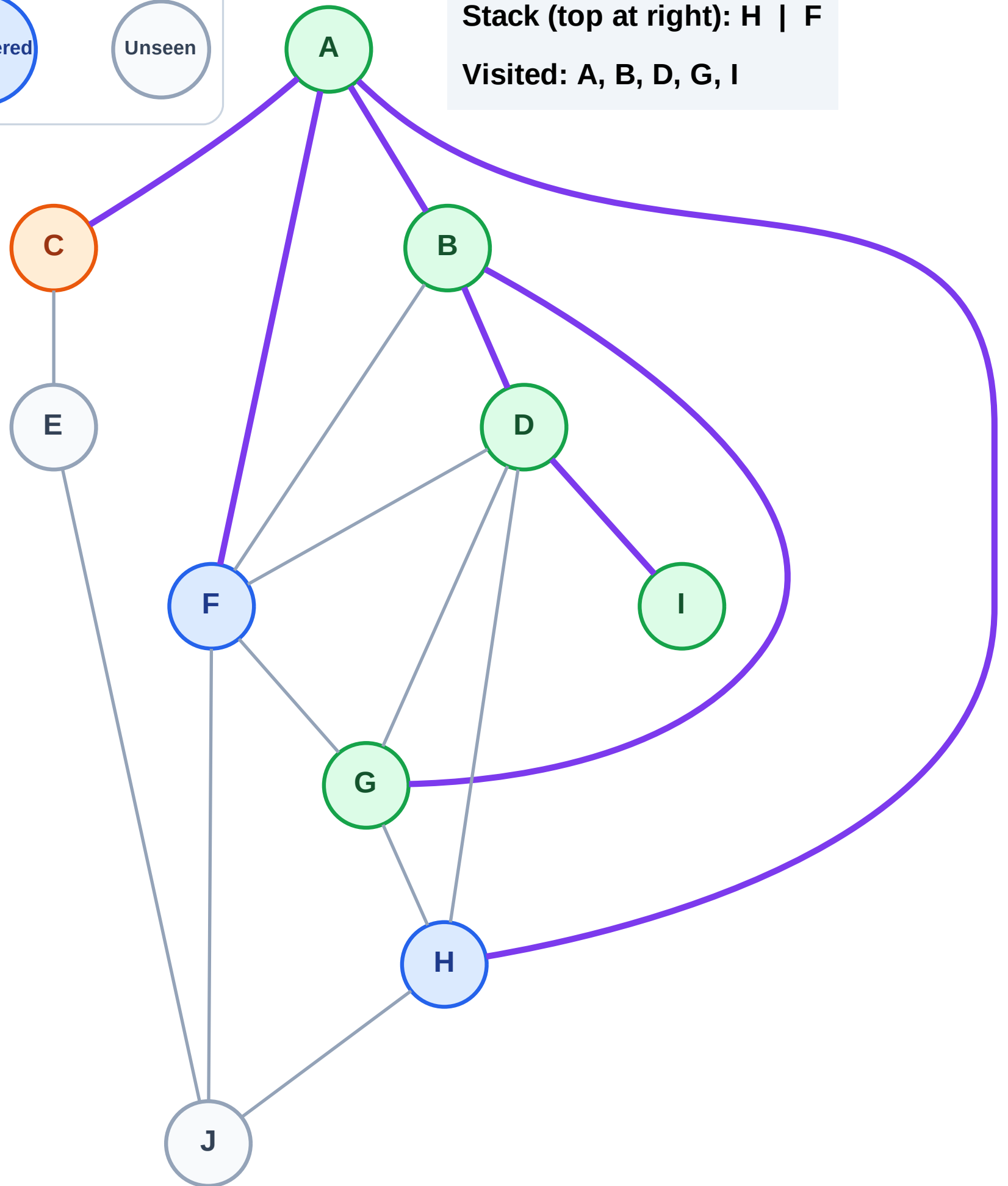
Discovered



Unseen

Stack (top at right): H | F

Visited: A, B, D, G, I



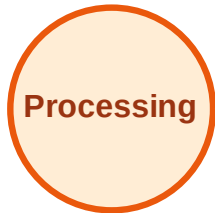
DFS Traversal

Step 13: Discover E from C

Legend



Complete



Processing



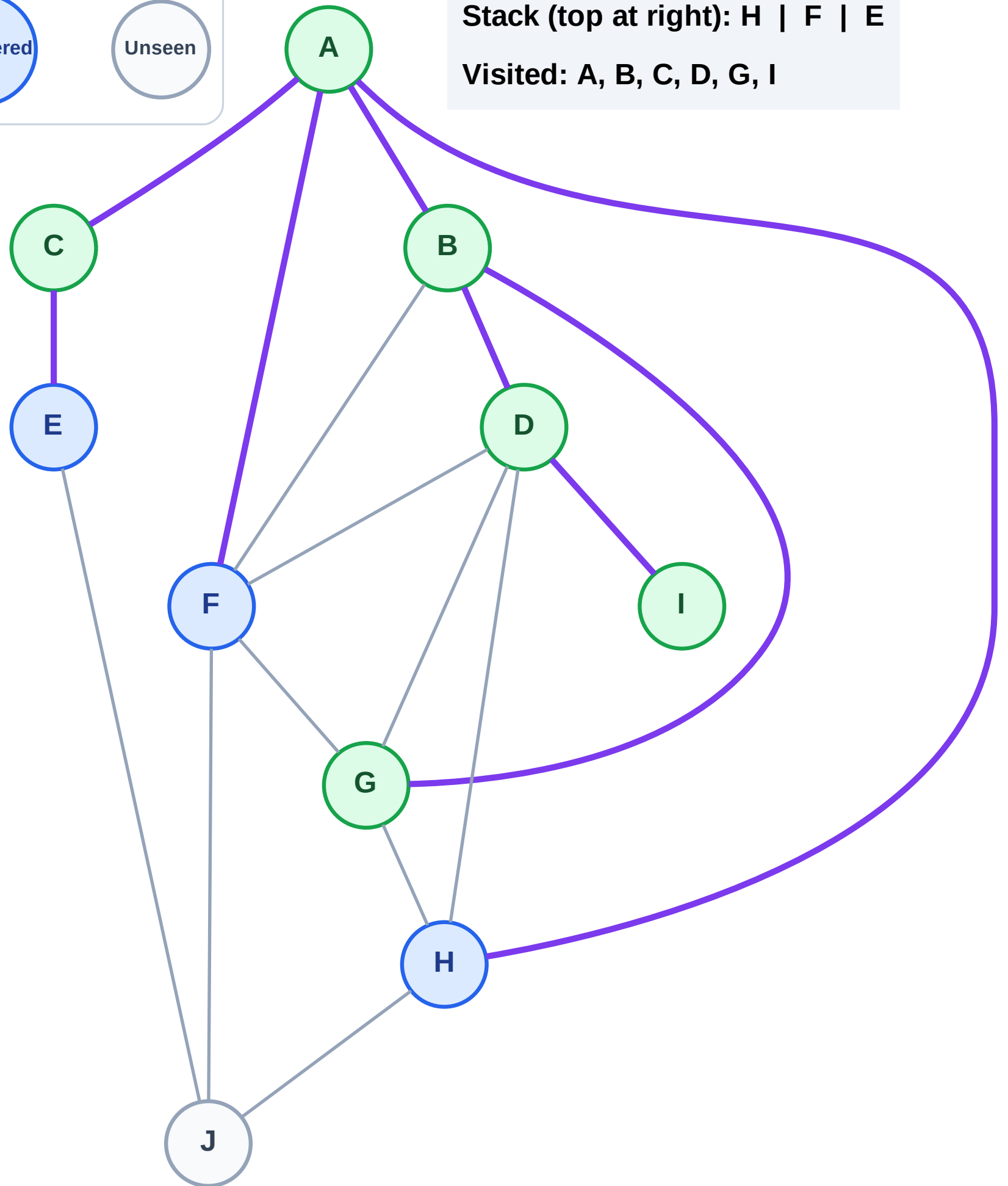
Discovered



Unseen

Stack (top at right): H | F | E

Visited: A, B, C, D, G, I



DFS Traversal

Step 14: Process node E

Legend

Complete

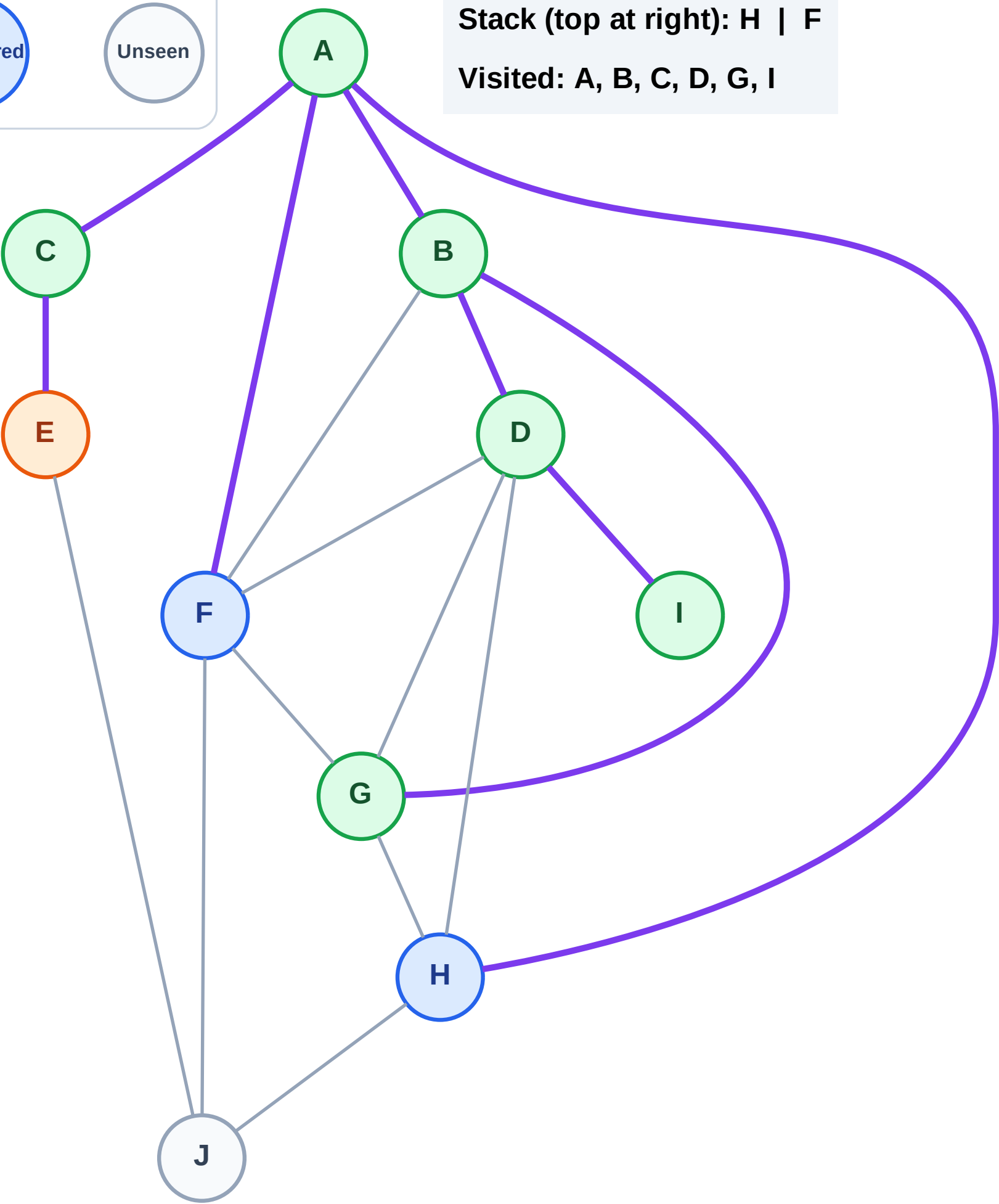
Processing

Discovered

Unseen

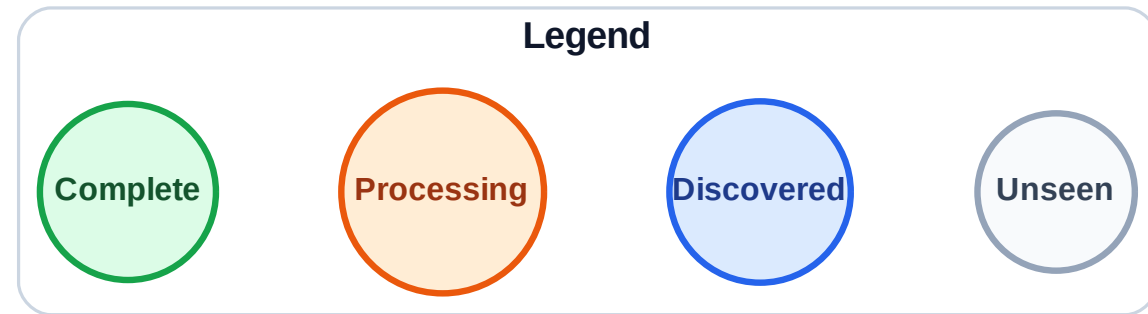
Stack (top at right): H | F

Visited: A, B, C, D, G, I

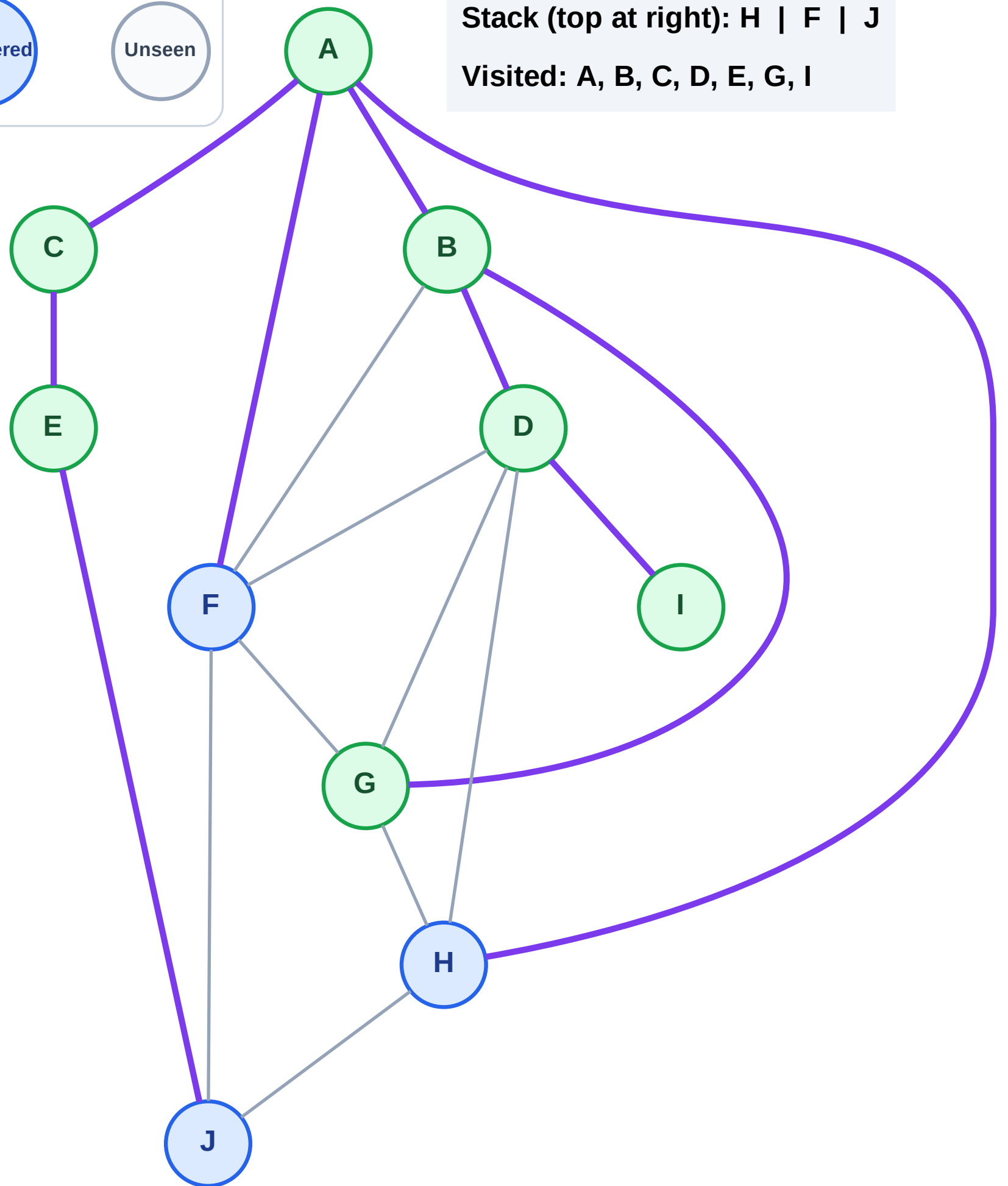


DFS Traversal

Step 15: Discover J from E



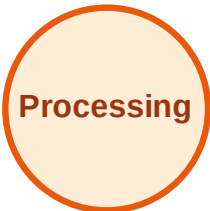
Stack (top at right): H | F | J
Visited: A, B, C, D, E, G, I



DFS Traversal

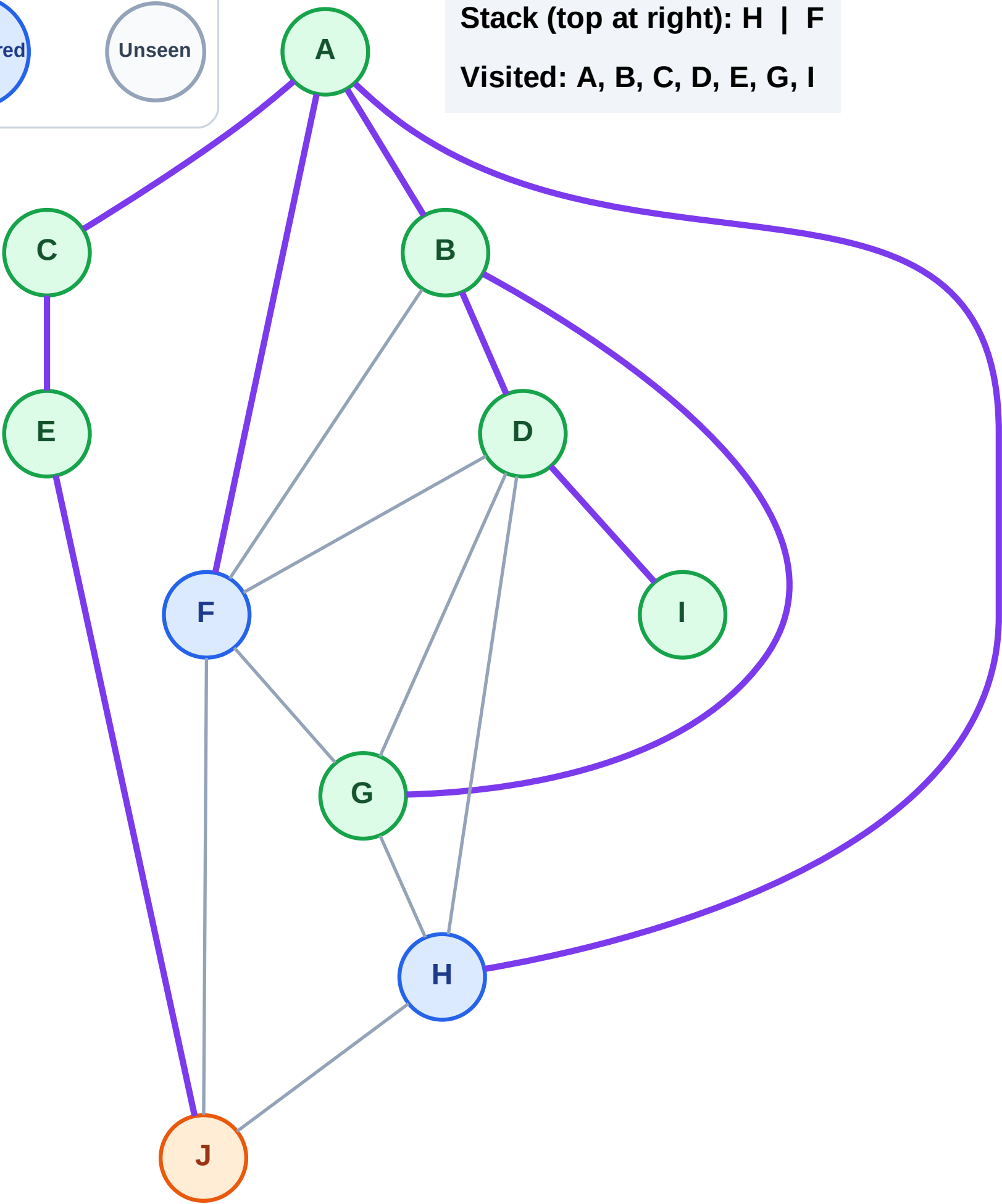
Step 16: Process node J

Legend



Stack (top at right): H | F

Visited: A, B, C, D, E, G, I



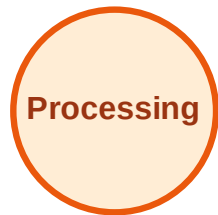
DFS Traversal

Step 17: No new neighbors from J

Legend



Complete



Processing



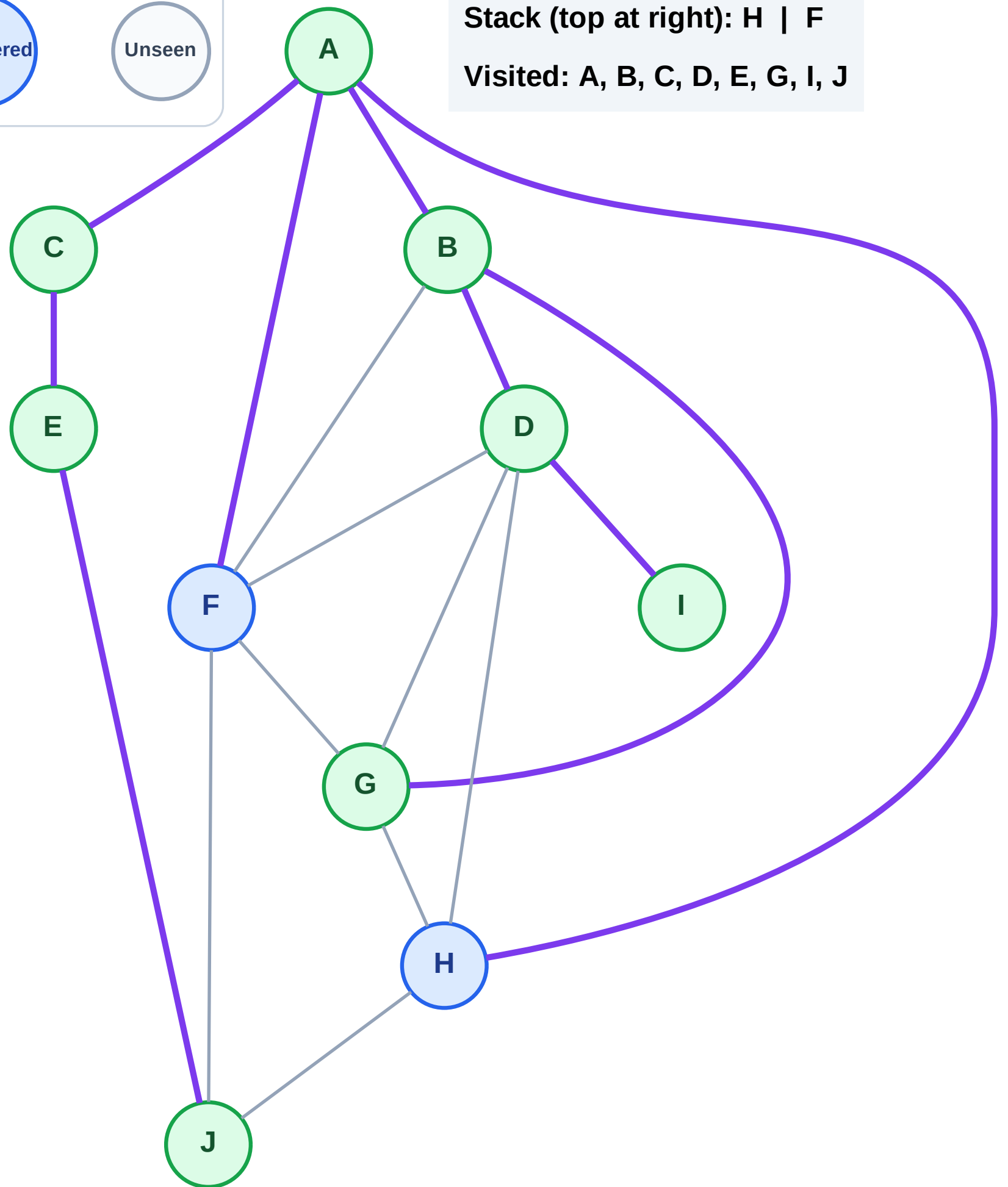
Discovered



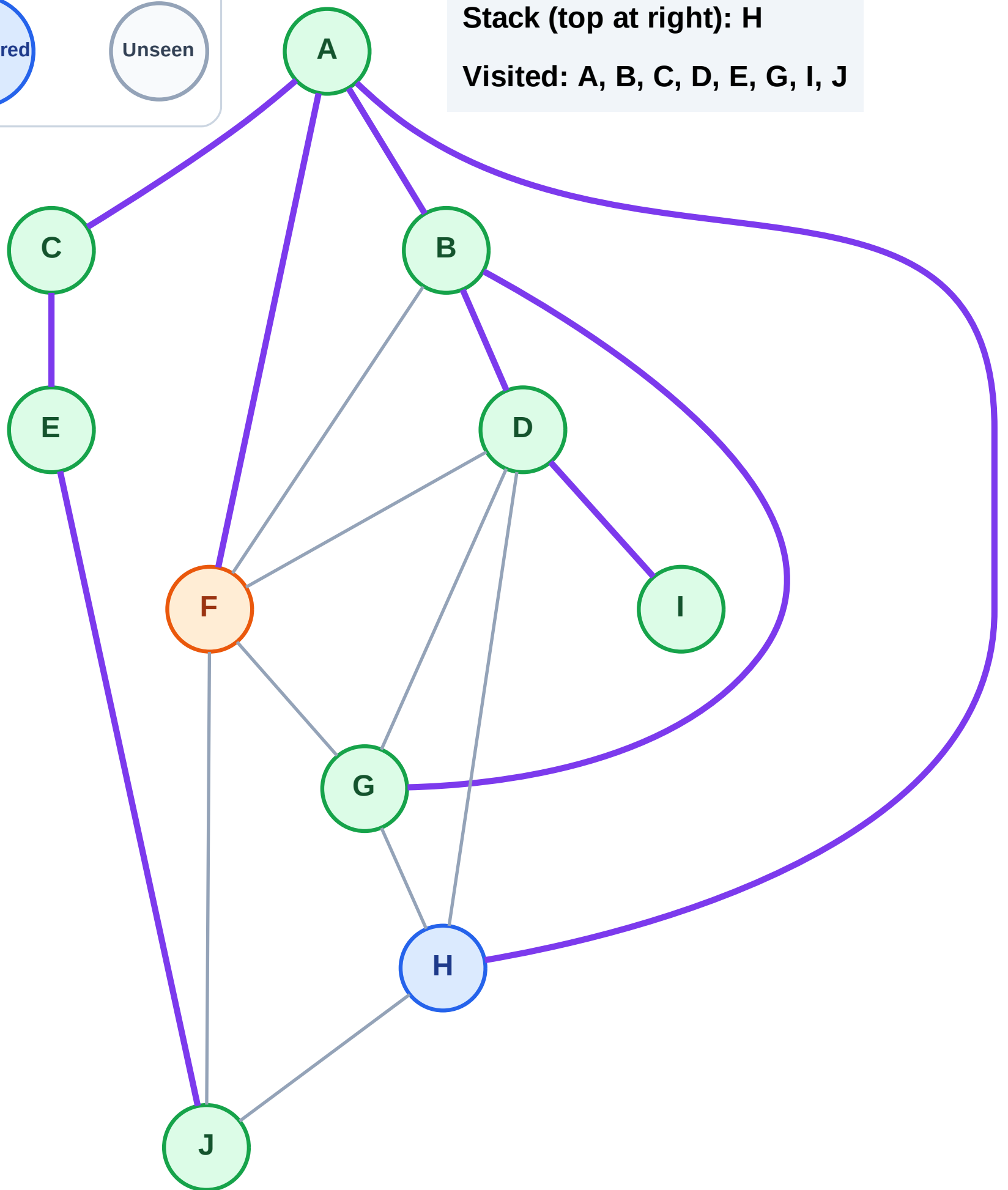
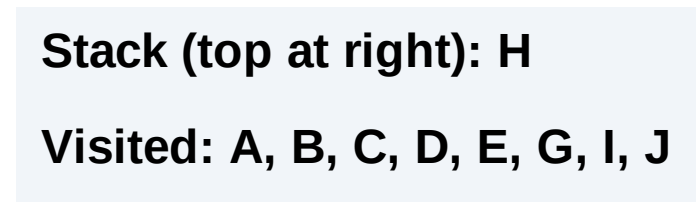
Unseen

Stack (top at right): H | F

Visited: A, B, C, D, E, G, I, J



Step 18: Process node F



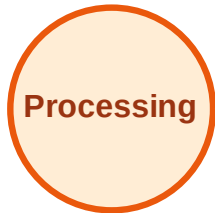
DFS Traversal

Step 19: No new neighbors from F

Legend



Complete



Processing



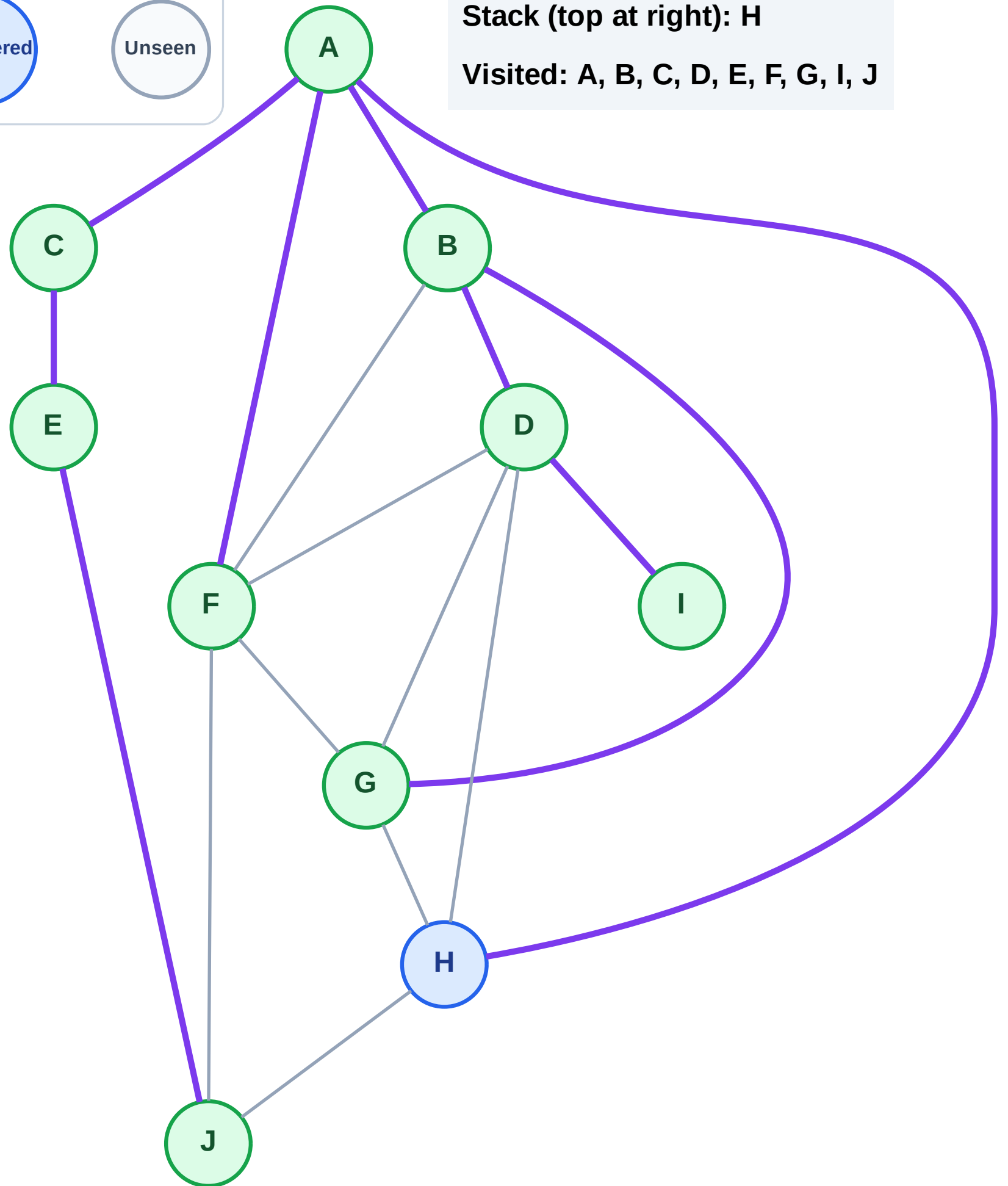
Discovered



Unseen

Stack (top at right): H

Visited: A, B, C, D, E, F, G, I, J



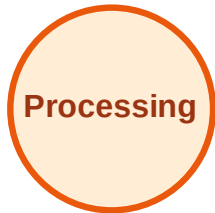
DFS Traversal

Step 20: Process node H

Legend



Complete



Processing



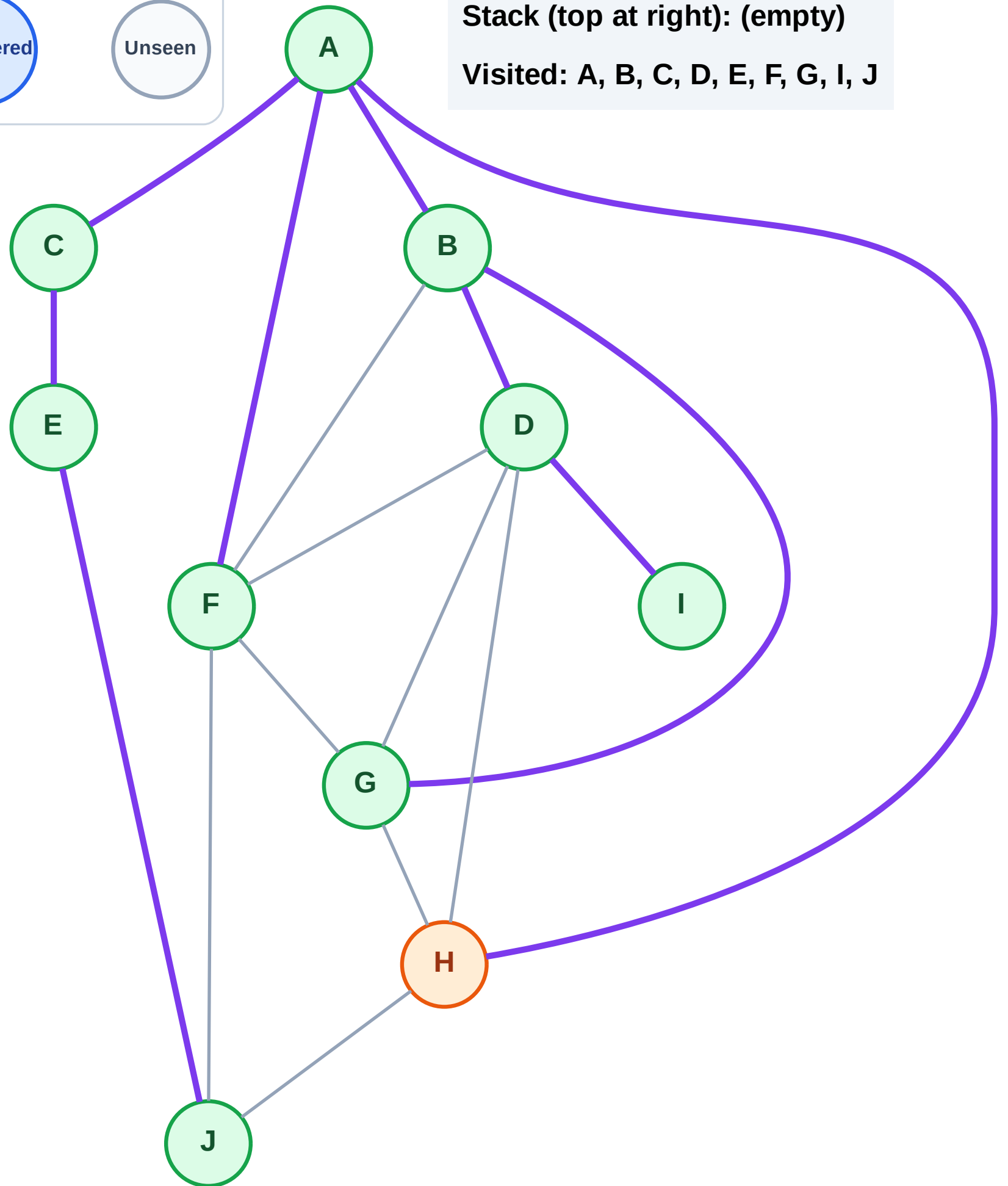
Discovered



Unseen

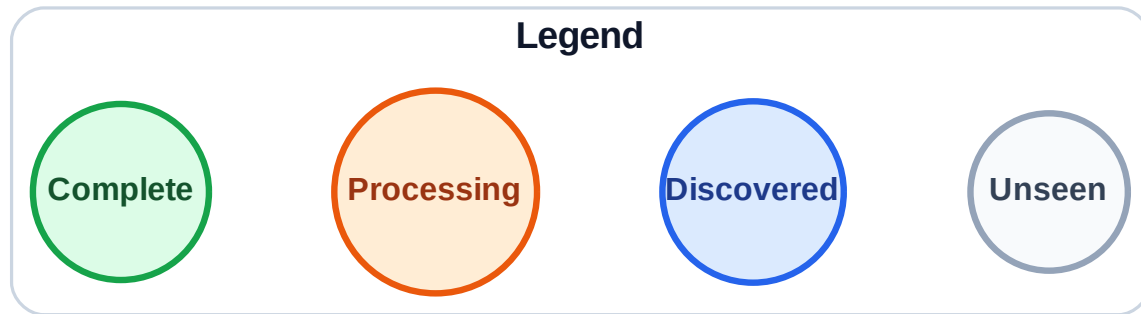
Stack (top at right): (empty)

Visited: A, B, C, D, E, F, G, I, J



Step 21: No new neighbors from H

Step 21: No new neighbors from H



Stack (top at right): (empty)

Visited: A, B, C, D, E, F, G, H, I, J

